Few-Shot Learning Based on Multilevel Contrast for Cross-Domain Hyperspectral Image Classification

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***Abstract*—Recently, domain adaptation techniques have been introduced for cross-domain few-shot hyperspectral image (HSI) classification tasks, but effectively aligning the source and target domains remains a core challenge. Current domain adaptation methods focus mainly on the instance level, which can easily damage the category separability of the target domain features. To address this, we propose a cross-domain multilevel contrast (CDMLC) method. We implement cross-domain few-shot learning (FSL) at the instance level, category level, and category distribution level. At the instance level, meta-learning is employed to capture meta-knowledge. At the category level, we design an intradomain contrastive loss with domain-mixed dictionaries. The intradomain contrastive loss can ensure feature compactness within each category in both domains and similar feature representations for semantically similar categories across domains. At the category distribution level, we perform category distribution fitting by supervised and unsupervised Gaussian mixture models (GMMs) in the source and target domains, respectively. Interdomain contrastive loss is constructed on the basis of the category distribution to promote domain alignment. Extensive experiments are conducted on five publicly available target HSI datasets, and the results demonstrate that our method outperforms the state-of-the-art FSL methods. The code is available at https://github.com/AAAA-CS/CDMLC**

***Index Terms*—Category prior, few-shot learning (FSL), meta-learning, multilevel contrast.**

I. Introduction

HYPERSPECTRAL images (HSIs) can be represented as 3-D data cubes that contain complex spatial information and redundant spectral information. HSIs have unique advantages for recognition and are widely used in the military and agriculture [1], [2], [3]. At present, HSIs provide potential advantages in land cover interpretation, and the HSI classification task has become a significant research focus in the field of remote sensing.

Early HSI classification methods focused primarily on traditional machine learning techniques such as support vector machines (SVMs) [4], k-nearest neighbors (KNNs) [5], random forests [6], and extended morphological profiles (EMPs) [7]. These feature extraction methods often rely on pixel-based statistical information or manually designed features to distinguish different land covers. Although traditional methods have been demonstrated to be effective, the quality of manually crafted features depends heavily on prior knowledge, resulting in lower classification accuracy and poorer robustness.

Owing to the significant advantages of deep neural networks (DNNs) in feature extraction, deep learning-based methods have been widely applied in HSI classification tasks [8], [9], [10]. The 2-D convolutional neural networks (2D-CNNs), 3-D convolutional neural networks (3D-CNNs), graph convolutional networks (GCNs), Transformer networks, and other advanced networks have been successfully employed. However, owing to the large number of spectral bands in HSIs and the high correlation between spectral bands, deep learning-based methods require many training samples to achieve ideal classification performance. The discrepancy between the scarcity of labeled samples and the requirement for many labeled samples severely limits the performance of deep learning-based models. Therefore, the few-shot problem is one of the primary challenges for improving the classification performance of deep learning-based HSI classification methods.

Cross-domain learning is a promising method for addressing the few-shot problem, which involves transferring knowledge from a source domain HSI dataset with sufficient labeled samples to a different target domain HSI dataset with few labeled samples. The typical cross-domain learning methods include domain adaptation [11], [12], [13] and meta-learning [14], [15]. Conventionally, domain adaptation requires that the source and target domains share the same semantic categories. However, in the study of HSI classification, which often focuses on different fine-grained categories across different domains due to significantly different acquisition conditions, domain adaptation is unsuitable for such scenarios. Therefore, we investigate the meta-learning paradigm to address this challenge [16], [17], [18]. Owing to the concept of meta-

Received 25 March 2025; revised 27 May 2025 and 27 June 2025; accepted 11 August 2025. Date of publication 18 August 2025; date of current version 4 September 2025. This work was supported in part by the Provincial Key Research and Development Program of Shaanxi under Grant 2024GH-ZDXM-47, in part by the Technology Innovation Leading Program of Shaanxi under Grant 2025QCY-KXJ-064, in part by Xi'an University of Technology International Science and Technology Cooperation Promotion Program under Grant 2024GHCJ021, and in part by the Education Department of Shaanxi Province Key Scientific Research Project under Grant 24JR123. *(Corresponding author: Li Fang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3599647

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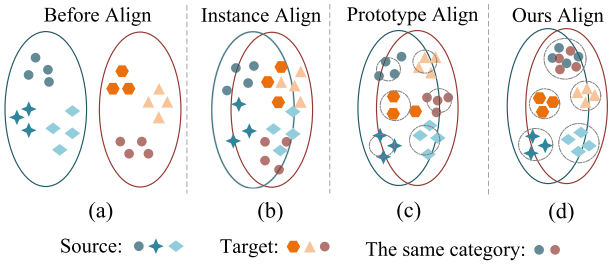


Fig. 1. (a) Data distributions of the source domain and target domain before domain alignment. There is a significant domain gap between the source and target domains. (b) Instance-level domain alignment. This method focuses solely on the alignment of global feature distributions while neglecting class-level structural information. (c) Category-prototype domain alignment. When the number of labeled samples is very limited, the class prototypes may fail to adequately represent the overall category distributions. Additionally, this approach often overlooks potential semantic category similarities between the source and target domains. (d) Our domain alignment. The category distributions of the source and target domains are better represented, and the semantic information is incorporated.

tasks, meta-learning has demonstrated superior performance in handling source and target domains with different categories. However, practical applications often involve significant differences in data distribution between the source and target domains, which reduce the transferability performance and classification accuracy.

To overcome the domain gap problem, various domain alignment methods have been introduced into few-shot learning (FSL). The typical domain alignment methods include adversarial learning-based methods [19], [20], [21], [22] and metric learning-based methods [23], [24], [25]. The former uses a domain discriminator to encourage the features from the source and target domains to be indistinguishable. The latter minimizes the metric distance between the feature distributions of the two different domains.

While the abovementioned methods have made progress in addressing domain alignment challenges, they still face significant limitations when applied to HSI analysis.

1) *Failure to Effectively Preserve Intraclass Compactness and Interclass Separability During Domain Alignment:* Hyperspectral data, characterized by large-scale spatial coverage and complex land cover characteristics, present unique challenges such as spectral variations among similar materials and material variations with similar spectral signatures. These complexities necessitate domain adaptation methods that go beyond simple instance-level alignment. Current approaches primarily reduce the domain shift by minimizing the feature distance between each pair of samples [23]. This direct alignment strategy fails to preserve intraclass compactness and interclass separability of target domain features, often leading to negative transfer, as shown in Fig. 1(b). While recent studies have attempted to address this limitation by incorporating class prototypes [26], [27], these solutions remain insufficient when dealing with the limited labeled samples typical of hyperspectral target domains. As shown in Fig. 1(c), the inherent diversity of hyperspectral data often exceeds the representational capacity of class prototypes derived from few-shot scenarios, resulting in suboptimal domain alignment.
2) *Ignoring the Semantic Category Similarity Between the Source and Target Domains:* Furthermore, existing methods overlook a critical aspect of hyperspectral data analysis: the semantic category similarity between source and target domains. In HSIs, the source and target domains typically share the same semantic categories, e.g., water and roads. The same or similar semantic categories should have similar representations to achieve effective domain alignment. However, current approaches fail to exploit this valuable information, potentially missing opportunities for more effective domain alignment in hyperspectral data analysis.

In this article, we explore the category characteristics within the meta-learning paradigm for cross-domain few-shot HSI classification. We propose a multilevel contrastive framework. Based on this framework, we propose a method that includes instance-level FSL, intradomain category-level feature learning and interdomain category-distribution-level alignment. For intradomain category-level feature learning, we construct a category-guided domain-mixed dictionary by utilizing intradomain category label information and interdomain semantic category information. On the basis of the category-guided domain-mixed dictionary, we design an intradomain category-level contrastive loss that forces feature compactness within each category in both domains while encouraging similar feature representations for semantically similar categories across domains. For interdomain category-distribution-level alignment, we use supervised Gaussian mixture models (GMMs) to cluster the category distribution of labeled samples in the source domain. To fully utilize both labeled and unlabeled samples in the target domain [28], we employ unsupervised GMMs to approximate the category distribution of unlabeled samples. This approach not only reduces the reliance on labeled data but also enhances the robustness of domain alignment in scenarios with limited annotations. On the basis of the category-level distribution, we design an interdomain category-distribution-level contrastive loss to align the compact category-level distribution and promote intercategory separability. The main contributions of this article are summarized as follows.

1) We propose an intradomain category-level contrastive learning framework. By incorporating a domain-mixed dictionary with category priors and an intradomain contrastive loss, our method explicitly maintains the compactness of category-specific features within both domains. Additionally, our method can ensure that semantically similar categories preserve feature consistency during the alignment process.
2) To address the limitations of class prototypes in capturing the diversity of hyperspectral data, we propose a category distribution estimation method using GMMs. This method provides more accurate category-level domain distributions while effectively incorporating unlabeled target domain samples, thereby significantly enhancing the completeness and robustness of target domain alignment in few-shot settings.

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3) Building on the GMM-based distribution estimation, we design an interdomain contrastive loss that aligns the GMM distribution (mean and variance) of matched categories across domains. This ensures that category semantics are preserved during alignment while maintaining intraclass compactness and interclass separability.

The remainder of this article is organized as follows. Section II covers the related work. Section III details the implementation of the proposed method. Section IV presents the experimental results and analysis. Section V provides the conclusion and future work of this article.

II. Related Works

A. DNN-Based HSI Classification Methods

There are three types of DNN-based HSI classification methods: spectral-, spatial-, and spectral–spatial-based methods. Current mainstream HSI classification methods utilize both spectral and spatial features for classification. In accordance with the structure of networks, we investigate the DNN-based HSI classification methods on the basis of local feature learning-based methods, global feature learning-based methods, and sequential feature learning-based methods.

1) Local Feature Learning-Based Methods: Local feature learning-based methods focus on capturing the spectral and spatial features at the pixel level. Song et al. [29] introduced residual learning into the HSI classification task, which addresses issues such as vanishing gradients and accuracy degradation. Guo et al. [30] integrated spatial and spectral attention mechanisms with a feature grouping strategy, which can effectively leverage the redundancy and diversity among spectral bands. Duan et al. [31] employed preprocessing feature extraction and postprocessing optimization to construct dual spatial feature extraction networks to facilitate the precise spectral–spatial feature extraction of HSIs. Chen et al. [32] combined local binary patterns and principal component analysis (PCA) to extract local spatial texture features, thereby enhancing the accuracy and generalization ability of the HSI classification model.

2) Global Feature Learning-Based Methods: Global feature learning-based methods consider the global spatial and spectral characteristics of HSIs, and graph neural networks (GNNs) are the primary feature extraction tool. Hong et al. [33] conducted qualitative and quantitative studies on convolutional neural networks (CNNs) and GCNs in HSI classification. They integrated the spectral–spatial features extracted by CNNs with the graph features for global feature learning. Yang et al. [34] designed spatial GCNs and spectral GCNs that incorporate attention mechanisms into graph information aggregation to construct adaptive cross-attention-driven spatial–spectral GCNs. To fully leverage unlabeled samples in HSIs, Mou et al. [35] constructed semisupervised nonlocal GCNs and represented entire HSIs as a nonlocal graphs.

3) Sequential Feature Learning-Based Methods: Sequential feature learning-based methods integrate both spatial information and spectral information sequentially. These methods aim to capture spatial–spectral dependencies across neighboring pixels and spectral bands. Yang et al. [36] designed HSI Transformer networks that embed convolution operations into the Transformer structure to capture spectral variations and local spatial contextual information of HSIs. Qing et al. [37] designed an end-to-end Transformer model (SATNet) that integrates spectral attention mechanisms and self-attention mechanisms for accurate sequential feature learning. He et al. [38] formulated the spectral–spatial Transformer framework, which leverages CNNs to extract spatial features and employs a dense connection-based Transformer to capture spectral sequence relationships. Li et al. [39] proposed an asymmetric encoder-decoder architecture called spectral coordinate Transformer (SCFormer) that takes into account the prior information of spectral coordinates. Luo et al. [40] proposed a self-supervised dual-structure Transformer for HSI clustering that improves the robustness and adaptability of the model when dealing with large-scale HSI datasets.

The abovementioned methods contribute collectively to the progress in HSI classification. However, these methods are susceptible to overfitting or underfitting with limited labeled samples.

B. Cross-Domain FSL

To address the challenge of cross-domain FSL, researchers have incorporated domain alignment strategies into few-shot HSI classification algorithms to reduce the domain shift, which include two main types: adversarial learning-based strategies and metric learning-based strategies.

1) Adversarial Learning-Based Cross-Domain FSL: Adversarial learning-based methods are based mainly on the concept of generative adversarial networks (GANs) [41]. Li et al. [42] introduced a conditional adversarial strategy into a deep FSL (DFSL) network [16] and constructed a deep cross-domain FSL (DCFSL) network. The framework unifies FSL and domain adaptation for the first time. Ye et al. [43] proposed adaptive domain adversarial FSL (ADAFSL) networks. The varying degrees of conditional distribution differences across different classes in the two domains are considered. Zhang et al. [44] divided the feature extraction layers into four blocks and established a domain discriminator for each block. This method aligns the data distributions of the two domains at different feature depths while preserving domain discriminative information. Liu et al. [20] designed domain discrimination and domain alignment branches for intermediate and output features, respectively, by using two conditional adversarial modules. Liu et al. [45] adopted a domain discriminator to address domain-level distribution differences while introducing contrastive learning to learn class-level sample relations.

These methods primarily depend on domain discriminators and conditional adversarial strategies to extract domain-invariant features. While these methods effectively address instance-level distribution discrepancies, they often compromise category separability of the target domain. Unlike these approaches, our method considers category-level domain alignment, which is a good strategy for preserving category separability during domain alignment.

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2) Metric Learning-Based Cross-Domain FSL: Cross-domain FSL always uses metric criteria to minimize the difference between the distributions of the domains. Zhang et al. [23] designed a graph information aggregation cross-domain FSL network (Gia-CFSL). It performs domain alignment on the feature maps and distribution maps via the graph optimal transport (GOT) and maximum mean discrepancy (MMD). Zhang et al. [24] obtained spatial features by stacking cross-modules and the graph sample and aggregation (Graph-SAGE) framework and then used the geodesic Wasserstein distance (GWD) to measure the spatial distribution differences between the two domains. Hu et al. [25] directly calculated the MMD value between the features of the source and target domains. They aligned the different domains by minimizing the MMD value. The work most related to our study is the work of Huang et al. [46], which proposed a domain adaptation method based on MoCo. However, it can achieve perfect performance only when the source and target domains have the same semantic categories, which is unrealistic for HSIs.

These methods generally measure the distance between feature distributions and achieve alignment by minimizing this distance. However, they similarly overlook the issue of maintaining interclass separability during the alignment process. Although some studies have recognized this problem and proposed prototype-based feature alignment approaches, the class prototypes estimated from a limited number of samples are unreliable. Therefore, we propose a category-distribution-level alignment model, which achieves reliable domain alignment and preserves intradomain feature separability by aligning class distributions.

C. Contrastive Learning

Recently, several instance contrastive learning methods have been proposed for learning discriminative feature spaces. InstDisc [47] initiated the memory bank-based contrastive paradigm by introducing an instance discrimination task with a memory bank. InvaSpread [48] built an end-to-end contrastive framework and proposed a compact sampling strategy that directly draws positive and negative pairs from the same mini-batch. Based on this, MoCo [49] improved the static memory bank of InstDisc to a dynamic queue structure and introduced a momentum encoder to ensure consistency in feature updates, effectively alleviating the limitation of negative sample size in end-to-end training. SimCLR [50] was built upon InvaSpread, enhanced the data augmentation pipeline, and systematically validated the synergistic effects of various augmentation strategies. Meanwhile, supervised contrastive learning [51] extended the contrastive learning paradigm to the supervised setting, explicitly promoting intraclass compactness and interclass separability using label information. In this article, on the basis of the MoCo dictionary creation and optimization strategy, we construct a domain-mixed dictionary and further construct a category-level contrastive strategy for domain alignment.

III. Method

The proposed cross-domain multilevel contrastive (CDMLC) method includes four modules: an embedding feature extraction module, an instance-level FSL module, an intradomain category-level feature learning module, and an interdomain category-distribution-level alignment module. The embedding feature extraction module extracts the domain invariant feature from two domains. The instance-level FSL module leverages a meta-learning strategy to acquire prior knowledge from the source domain and discriminative knowledge from the target domain. The intradomain category-level feature learning module enhances the category discriminability within each domain and the interdomain semantic consistency via contrastive learning with category constraints. The interdomain category-distribution-level alignment module aligns the category-level distribution via a supervised GMMs in the source domain and an unsupervised GMMs in the target domain. The overall framework of the CDMLC method is shown in Fig. 2. In the following, we provide a detailed introduction to each part.

A. Embedding Feature Extraction

In this work, we utilize the 3D-CNN and residual structure for embedding feature extraction, which can simultaneously extract the spectral and spatial features of HSIs. The structure of the embedding feature extractor, which consists of a mapping module and two residual blocks, is shown in Fig. 3. The mapping module consists of a 2-D convolutional layer and a 2-D batch normalization layer. The purpose of the mapping module is to align the spectral dimensions of the source and target domains to be the same. Each residual block is composed of three 3-D convolutional layers with skip connections between the first and third convolutional layers. The residual blocks are connected by a max pooling layer. The output of the residual blocks, after max pooling, is then passed through a 3-D convolution layer and a linear layer to obtain the final embedded features. The embedding feature extractor is used to extract domain-invariant spatial–spectral embedding features.

B. Instance-Level FSL With Meta-Learning

Meta-learning involves the collection of numerous meta-tasks known as episodes from the source domain to emulate FSL tasks. After being trained using the episodic paradigm, the model is capable of extracting general patterns from a diverse set of labeled samples in the source domain. When confronted with novel categories in the target domain, it can utilize the meta-knowledge obtained from the source domain to achieve precise classification in the target domain. Therefore, our CDMLC method adopts a meta-learning framework for cross-domain FSL.

We assume that the source domain HSI dataset is D^s with C^s categories and that the target domain HSI dataset is D^t with C^t categories. Compared with the target domain, the source domain contains an equal or greater number of categories, i.e., $C^s \geq C^t$. In this article, we introduce our method by assuming that the source and target domains contain the same semantic categories, and the case where they do not share the same semantic categories is an exception. In the training process of meta-learning, for each episode, N categories are

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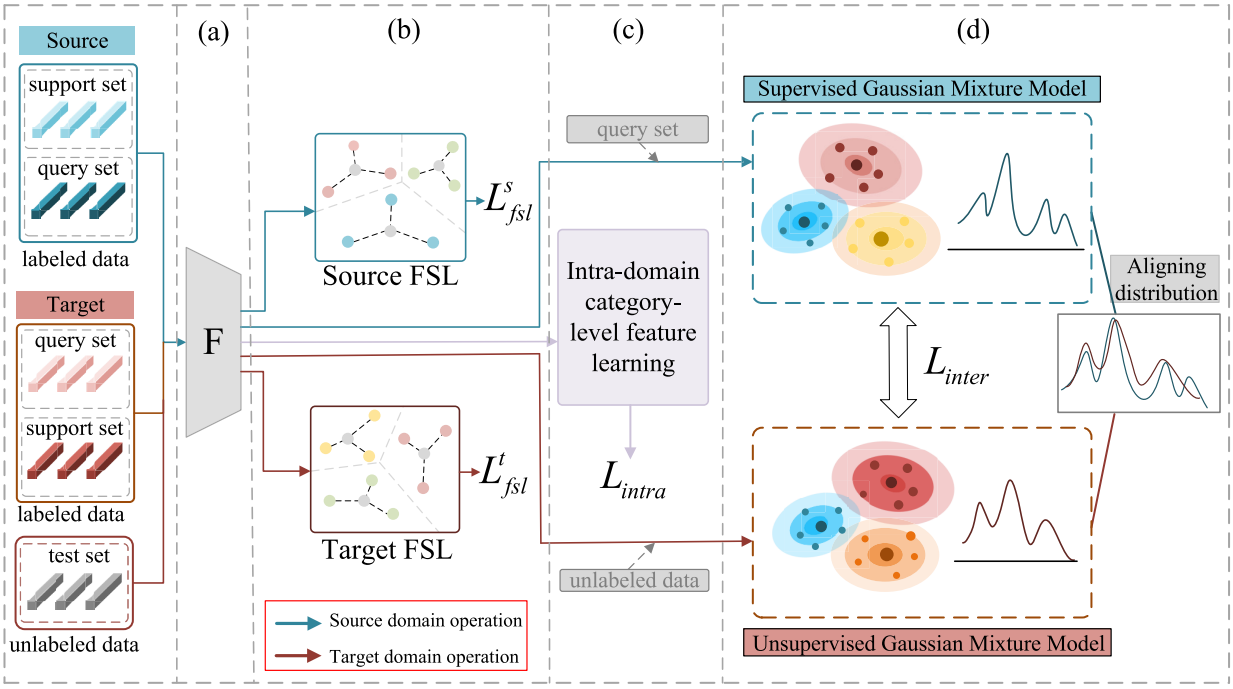


Fig. 2. Overall framework of the CDMLC framework. The CDMLC framework comprises the following four parts. (a) Embedding feature extraction, (b) instance-level FSL, (c) intradomain category-level feature learning, and (d) interdomain category-distribution-level alignment.

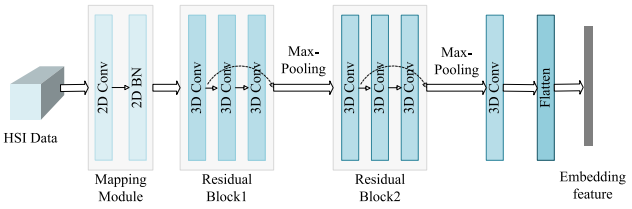


Fig. 3. Structure of the embedding feature extractor. Dashed lines represent skip connections.

randomly sampled from both the source and target domains, with K samples per category collected as the support set and Q samples per category as the query set, forming N-way K-shot Q-query episodes. In this article, the number of sample categories N is equal to the number of categories in the target domain, i.e., $C^s \geq C^t = N$. The support set is used to train the embedding feature extractor, and the query set is used to construct the loss function and update the parameters of the embedding feature extractor. Training is conducted iteratively on the data of the source and target domains.

The support set and query set of the source domain are denoted as $S_s = \{(x_{is}, y_{is}, \widehat{y}_{is})\}_{is=1}^{N\times K}$ and $Q_s = \{(x_{js}, y_{js}, \widehat{y}_{js})\}_{js=1}^{N\times Q}$, respectively. In the support set, $x_{is} \in R^{w\times w\times d}$ is the support sample, where $w \times w$ is the patch size and d is the number of bands for HSIs. $y_{is} \in \{1, \ldots, N\}$ is the sampled label, and $\widehat{y}_{is} \in \{1, \ldots, C^s\}$ is the true category label of sample x_{is} in the source domain. In the query set, $x_{js} \in R^{w\times w\times d}$ is the query sample, $y_{js} \in \{1, \ldots, N\}$ is the sampled label, and $\widehat{y}_{js} \in \{1, \ldots, C^s\}$ is the true category label of sample x_{js} in the source domain. After passing through the feature extractor, the query samples and support samples are projected into an embedding space. Based on the prototype networks [52], we construct class prototypes by utilizing the features of samples in the support set S_s, as expressed in the following:

$$P_{ns} = \frac{1}{K} \sum_{is \in S_s(y_{is}=n)} f_\varphi(x_{is}) \tag{1}$$

where f_φ is the embedding feature extractor and P_{ns} is the class prototype of the nth sampled category, which is obtained by averaging the features of all samples in the support set within the nth sampled category. In the following, we denote the embedding feature $f_\varphi(x_{is}) \in R^{\text{dim}}$ as $f_{x_{is}}$, where dim is the dimensions of the embedding feature.

For each query sample x_{js}, the distance between the feature and each class prototype in the support set is calculated, and the category with the highest probability is the label. The category distribution of the query sample x_{js} is shown as follows:

$$p_{\varphi s}(y_{js} = n|x_{js}) = \frac{\exp\left(-d\left(f_{x_{js}}, P_{ns}\right)\right)}{\sum_{n=1}^{N} \exp\left(-d\left(f_{x_{js}}, P_{ns}\right)\right)} \tag{2}$$

where $d(\cdot)$ represents the Euclidean distance and $p_{\varphi s}$ represents the probability of query sample x_{js} belonging to the nth sampled category. The instance-level FSL loss function of the source domain is shown as follows:

$$L_{fsl}^s = E_{S_s,Q_s}\left[-\sum_{(x_{js},y_{js})\in Q_s} \log p_{\varphi s}(y_{js} = n|x_{js})\right]. \tag{3}$$

We denote the support set and query set of the target domain as $S_t = \{(x_{it}, y_{it}, \widehat{y}_{it})\}_{it=1}^{N\times K}$ and $Q_t = \{(x_{jt}, y_{jt}, \widehat{y}_{jt})\}_{jt=1}^{N\times Q}$,

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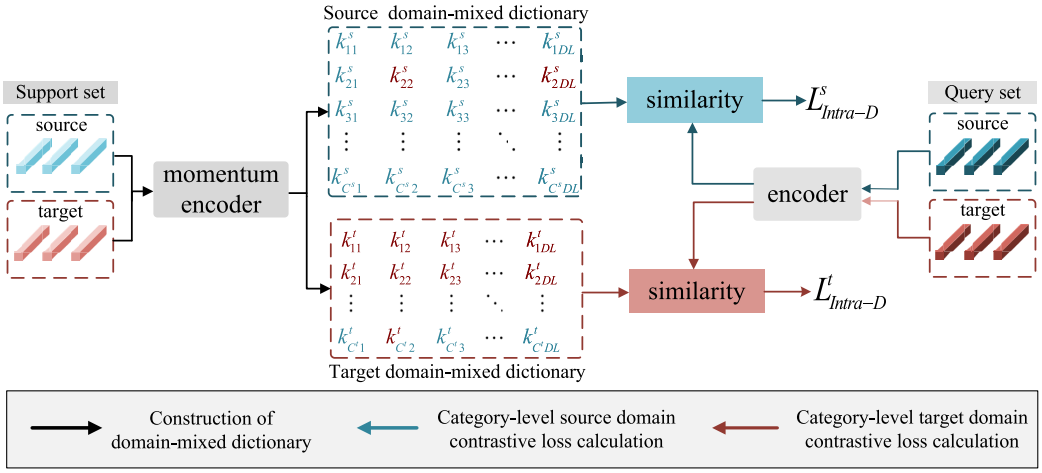


Fig. 4. Construction of the domain-mixed dictionary and the calculation process of the category-level contrastive loss for the source and target domains. The term "similarity" refers to a metric that quantifies the degree of resemblance between two samples in the feature space, which is computed using cosine similarity. The objective is to learn more discriminative feature representations by maximizing the similarity of positive sample pairs while minimizing the similarity of negative sample pairs.

respectively. The instance-level FSL loss function of the target domain is similarly defined as follows:

$$L^t_{fsl} = E_{S_t,Q_t}\left[-\sum_{(x_{jt},y_{jt})\in Q_t} \log p_{\varphi t}\left(y_{jt}=n|x_{jt}\right)\right] \tag{4}$$

where $p_{\varphi t}$ represents the probability of query sample x_{jt} belonging to the nth sampled category.

The total instance-level FSL loss function is defined as follows:

$$L_{fsl} = L^s_{fsl} + L^t_{fsl}. \tag{5}$$

C. Intradomain Category-Level Discriminative Feature Learning

Instance contrastive learning methods often lack a category prior; therefore, some negative keys have similar embedding features with positive keys, leading to reduced category discriminability. To learn more discriminative intradomain spatial–spectral feature representations, we construct category-guided domain-mixed dictionaries for the source and target domains and design a category-level contrastive loss to promote feature compactness within each category and feature similarity for semantically similar categories across different domains.

We construct the domain-mixed dictionaries using encoded support samples from episodes in both the source and target domains. The source domain-mixed dictionary is denoted as $dc^s = \{k^s_{hw}\}_{1\le h\le C^s, 1\le w\le \mathrm{DL}}$, where $k^s_{hw}=f^s_k(x_{js}) \in R^{\dim}$ is an encoded support sample and f^s_k is a source momentum encoder. In the source domain-mixed dictionary, the number of rows is the number of true categories in the source domain, and the number of columns is a predetermined dictionary length. Therefore, each line in the source domain-mixed dictionary represents a queue for each true category. In each episode, the encoded support samples are stored at the end of the corresponding category queues according to their true categories. Additionally, if the encoded support samples in the target domain have the same semantic categories as the source domain, the encoded support samples in the target domain are also stored at the end of the corresponding category queues according to their semantic categories. Here, we consider categories with identical semantic labels as the same semantic categories. During the subsequent meta-training process, the source domain-mixed dictionary is updated dynamically by adding the encoded support samples of the current episode and removing the earliest added encoded support samples if the queue is full. According to this process, we construct a dynamic source domain-mixed dictionary. In the target domain, the target domain-mixed dictionary is denoted as $dc^t = \{k^t_{hw}\}_{1\le h\le C^t, 1\le w\le \mathrm{DL}}$, where the number of rows is the number of true categories in the target domain and the number of columns is the same as that in the source domain-mixed dictionary. The target domain-mixed dictionary construction process is similar to that of the source domain-mixed dictionary and is also updated dynamically. The internal structure and computation process of the category-level discriminative feature learning module are shown in Fig. 4.

Since the encoded support samples stored in the dictionary are added from different episodes, the encoder features have inconsistent characteristics; therefore, we adopt the MoCo optimization strategy to achieve intradomain category-level contrastive learning on the basis of the domain-mixed dictionary.

For the source domain, we use the embedding feature extractor f_φ to extract the embedding feature of the query sample. In order to alleviate the inconsistency between the samples in the training process, we set the source domain encoder f^s_k as a momentum encoder with the same structure of f_φ to encode the support samples. The parameters of the source domain momentum encoder f^s_k are updated based on the parameters of the embedding feature networks f_φ, as shown as follows:

$$\theta_{f^s_k} = b\theta_{f^s_k} + (1-b)\,\theta_{f_\varphi} \tag{6}$$

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where b is a momentum coefficient and θ_{f_φ} and $\theta_{f_k^s}$ are the parameters of the embedding feature extractor and source domain momentum encoder, respectively. Then, in each episode, for a query sample x_{js}, the encoded positive and negative samples in the source domain-mixed dictionary are matched according to its true category. The category-level source domain contrastive loss is defined in (7), as shown at the bottom of the page, where $\widehat{y}_{k_{hw}^s}$ is the category label for sample k_{hw}^s in the source domain-mixed dictionary. $\mathbf{I}(\widehat{y}_{js} == \widehat{y}_{k_{hw}^s}) = 1$ if $\widehat{y}_{js}$ and $\widehat{y}_{k_{hw}^s}$ are the same category label; otherwise, it equals 0. $\mathbf{I}(\widehat{y}_{js} \neq \widehat{y}_{k_{hw}^s}) = 1$ if $\widehat{y}_{js}$ and $\widehat{y}_{k_{hw}^s}$ are different category labels; otherwise, it equals 0.

For the target domain, we also set a target domain momentum encoder f_k^t and define a category-level target domain contrastive loss in (8), as shown at the bottom of the page.

The overall intradomain contrastive loss is shown as follows:

$$L_{\text{Intra}} = L_{\text{Intra}-D}^s + L_{\text{Intra}-D}^t. \tag{9}$$

Intradomain category-level feature learning is implemented once the domain-mixed dictionaries of both the source and target domains are filled.

D. Interdomain Category-Distribution-Level Alignment

In this section, we implement domain alignment on the basis of compact intradomain feature distribution while ensuring interclass separability.

From the perspective of data distribution, for each class, many sample features are close to the class center, whereas a small number of sample features near the classification boundary are far from the class center. Therefore, the sample features within each category usually exhibit an approximate Gaussian distribution. Thus, we assume that the sample features within a class follow a Gaussian distribution and use the Gaussian distribution to achieve category-distribution-level domain alignment.

There are sufficient labeled samples in the source domain; therefore, we use the embedding features of the query samples in each episode to perform Gaussian fitting for each sampled category. The category-distribution-level Gaussian distribution is expressed as follows:

$$G_n^s = \phi\left(\left(y_{js} = n | f_{x_{js}}\right) | \mu_n^s, \sigma_n^s\right) \tag{10}$$

where $(y_{js} = n|f_{x_{js}})$ represents the embedding features of the query samples in the nth sampled category; $\mu_n^s \in R^{\text{dim}}$ and $\sigma_n^s \in R^{\text{dim}}$ represent the corresponding mean and variance values, respectively; $\phi(\cdot)$ is the Gaussian distribution function; and G_n^s is the Gaussian feature distribution of the nth sampled category in the source domain.

In the target domain, the number of labeled samples is limited; therefore, the predicted Gaussian distribution based on fewer labeled samples is unreliable and does not fully reflect the true category distribution. Hence, we consider the use of many unlabeled samples from the target domain to achieve reliable Gaussian fitting. We utilize unsupervised GMMs to cluster the unlabeled samples. Specifically, in each episode, we randomly select 1000 unlabeled samples from the target domain as fitting samples, with the number of clusters set to the number of categories in the target domain. Note that the number of categories in the target domain is the same as the number of sampled categories, i.e., $C^t = N$; thus, the source and target domains have the same number of clusters.

We assume that $\mathbf{x_t}$ represents all unlabeled samples to be fit in the target domain and that $x_{ut} \in \mathbf{x_t}$. $n, c \in \{1, 2, \ldots, N\}$ is the number of clusters. The unsupervised GMM can be defined as follows:

$$G^t = \sum\nolimits_{n=1}^{N} \alpha_n \varphi\left(f_{\mathbf{x_t}} | \mu_n^t, \sigma_n^t\right) \tag{11}$$

where μ_n^t and σ_n^t represent the mean and variance, respectively, of the embedding features in the nth sampled category in the target domain and α_n represents the mixture coefficient of the nth Gaussian distribution. According to (11), the posterior probability that is used to assign the embedding feature $f_{x_{ut}}$ to the nth Gaussian distribution is defined as follows:

$$\gamma_{f_{x_{ut}},n} = \frac{\alpha_n \varphi\left(f_{x_{ut}} | \mu_n^t, \sigma_n^t\right)}{\sum_{c=1}^{N} \alpha_c \varphi\left(f_{x_{ut}} | \mu_c^t, \sigma_c^t\right)} \tag{12}$$

where μ_n^t, σ_n^t, and α_n are determined via the expectation–maximization (EM) algorithm.

In the Gaussian distribution, the mean determines the positions of different categories in the decision space, whereas the variance represents the degree of data dispersion. In the following, we implement domain alignment on the basis of the obtained mean and variance distributions of each category. The schematic of mean distribution alignment and variance distribution alignment is shown in Fig. 5.

To ensure intradomain compactness and interdomain separability in domain alignment, we design a category-distribution-level alignment loss based on the mean and variance of each category, as shown as follows:

$$L_{\text{Inter}} = L_\mu + L_\sigma \tag{13}$$

$$L_{\text{Intra}-D}^s = \sum_{js=1}^{N\times Q} -\log \frac{\sum_{h=1,w0=1}^{h=C^s,w=\text{DL}} \mathbf{I}\left(\widehat{y}_{js} == \widehat{y}_{k_{hw}^s}\right) \exp\left(f_{x_{js}} \cdot k_{hw}^s/\tau\right)}{\sum_{h=1,w=1}^{h=C^s,w=\text{DL}} \mathbf{I}\left(\widehat{y}_{js} \neq \widehat{y}_{k_{hw}^s}\right) \exp\left(f_{x_{js}} \cdot k_{hw}^s/\tau\right)} \tag{7}$$

$$L_{\text{Intra}-D}^t = \sum_{jt=1}^{N\times Q} -\log \frac{\sum_{h=1,w=1}^{h=C^t,w=\text{DL}} \mathbf{I}\left(\widehat{y}_{jt} == \widehat{y}_{k_{hw}^t}\right) \exp\left(f_{x_{jt}} \cdot k_{hw}^t/\tau\right)}{\sum_{h=1,w=1}^{h=C^t,w=\text{DL}} \mathbf{I}\left(\widehat{y}_{jt} \neq \widehat{y}_{k_{hw}^t}\right) \exp\left(f_{x_{jt}} \cdot k_{hw}^t/\tau\right)} \tag{8}$$

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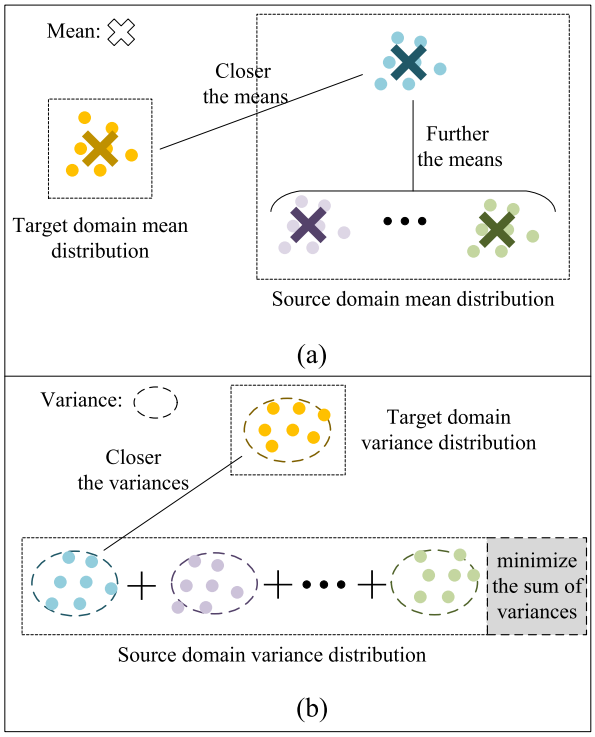


Fig. 5. Framework of interdomain category-distribution-level alignment. Taking the Gaussian distribution of a class in the target domain as an example. (a) Mean distribution alignment. (b) Variance distribution alignment.

where L_μ and L_σ are the mean distribution alignment loss and variance distribution alignment loss, respectively,

$$L_\mu = \sum_{\mu_i^s \in D^s} -\log \frac{\sum_{j=1}^{N} I\left(\mu_i^s == \mu_j^t\right) \exp\left(\mu_i^s \cdot \mu_j^t / \tau\right)}{\sum_{k=1}^{N} I\left(i \neq k\right) \exp\left(\mu_i^s \cdot \mu_k^s / \tau\right)} + \sum_{\mu_j^t \in D^t} -\log \frac{\sum_{i=1}^{N} I\left(\mu_j^t == \mu_i^s\right) \exp\left(\mu_j^t \cdot \mu_i^s / \tau\right)}{\sum_{k=1}^{N} I\left(j \neq k\right) \exp\left(\mu_j^t \cdot \mu_k^t / \tau\right)} \tag{14}$$

$$L_\sigma = \sum_{\sigma_i^s \in D^s} \sum_{j=1}^{N} I\left(\sigma_i^s == \sigma_j^t\right) d\left(\sigma_i^s, \sigma_j^t\right) + \sum_{\sigma_i^s \in D^s} \sigma_i^s + \sum_{\sigma_j^t \in D^t} \sigma_j^t. \tag{15}$$

In (14), $I(\mu_i^s == \mu_j^t) = 1$ if μ_i^s and μ_j^t are the matched positive keys of the source and target domains; otherwise, it equals 0. $I(i \neq k) = 1$ if the index values of i and k are different; otherwise, it equals 0. The first term indicates that the ith mean distribution in the source domain is close to the jth matched mean distribution in the target domain but distant from the other mean distributions in the source domain. The second term indicates that the jth mean distribution in the target domain is close to the ith mean distribution in the source domain but distant from the other mean distributions in the target domain.

In (15), $I(\sigma_i^s == \sigma_j^t) = 1$ if σ_i^s and σ_j^t are the matched positive keys of the source and target domains; otherwise, it equals 0. The first term minimizes the distance of the matched positive keys. The second and third terms minimize the sum of the variances for each category in the source and target domains, respectively.

For the construction of positive and negative sample pairs, take the source domain as an example, and the construction of positive sample pairs begins by determining whether the current source domain category distribution shares the same semantic categories with the target domain category distribution. If the source and target domains share the same semantic categories, they are directly matched as positive pairs (based on the assumption that features of the same category should be close to each other). Negative samples are defined as other category distributions from the source domain. For the remaining source domain category distributions, we select sequentially them as anchors and randomly select category distributions from the remaining target domain to form matched positive pairs. For the target domain, we similarly select sequentially category distributions as anchors, with positive sample pairs matched as described above and negative sample pairs defined as other category distributions within the target domain.

Through this contrastive learning process, the category distributions of the source and target domains are brought closer together, while the interclass discrepancies within each domain are enlarged to maintain category discriminability. This approach ensures that the alignment process is both effective and semantically meaningful, even in scenarios where semantic categories may not fully overlap.

In conclusion, the mean distribution alignment loss aligns the feature cluster centers of different domains while encouraging the feature cluster centers of different categories within the domain to be far away. The variance distribution alignment loss ensures that the feature distribution of each category has a smaller variance, which can reduce the number of confusing samples near the classification boundary after domain alignment. Thus, category-distribution-level alignment loss achieves discriminative domain alignment from the perspective of category distributions.

E. Overall Training of the CDMLC Framework

By considering instance-level FSL loss, intradomain category-level contrastive loss, and interdomain category-distribution-level alignment loss, the total loss is shown as follows:

$$L = L_{fsl} + \beta L_{\text{Intra}} + \gamma L_{\text{Inter}}. \tag{16}$$

The loss functions of the source and target domains are calculated iteratively. More specifically, the training procedure of CDMLC is available in Algorithm 1. In the testing phase, the training and testing samples from the target domain are input into the pretrained feature extractor. The classification results for each sample are then obtained using the KNN algorithm.

IV. Experiments

All experiments were conducted on a desktop equipped with an Intel Core i7-11700K CPU (3.6 GHz), 32 GB of memory, and an Nvidia GeForce RTX 3090 GPU. The training and testing procedures were implemented using the open-source software framework PyTorch.

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Algorithm 1 Training Process of CDMLC

Input: S_s, Q_s of D^s; S_t, Q_t of D^t; D^t; the number of training episodes

Output: The classification map of the target dataset

Initialize parameters f_φ
Initialize parameters f_k^s, f_k^t by f_φ
for episode = 1 **to** episodes **do**
1. Instance-level FSL
Compute embedding features $f_{x_{is}}$, $f_{x_{js}}$, $f_{x_{it}}$, $f_{x_{jt}}$ via f_φ
Compute class prototypes by Eq.1
Compute category distribution of query samples by Eq.2
Compute the loss L_{fsl} by Eq.3, Eq.4 and Eq.5
2. Intradomain category-level discriminative feature learning
Compute embedding features $f^s_{k(x_{is})}$, $f^t_{k(x_{it})}$ via f_k^s, f_k^t
Construct domain-mixed dictionaries dc^s, dc^t using $f^s_{k(x_{is})}$, $f^t_{k(x_{it})}$ and $\widehat{y}_{is}$, $\widehat{y}_{it}$
Compute loss L_{Intra} via Eq.7, Eq.8 and Eq.9
3. Interdomain category-distribution-level alignment
Perform Gaussian fitting for $f_{x_{js}}$ via Eq.10
Randomly select 1000 unlabelled samples $\mathbf{x_t}$ from D_t
Perform Gaussian fitting for $f_{\mathbf{x_t}}$ via Eq.11 and Eq.12
Construct positive and negative sample pairs for class distributions
Compute loss L_{Inter} via Eq.13
Compute total loss $\mathcal{L}$ via Eq. 16
Update parameters f_φ by minimizing $\mathcal{L}$
Update parameters f_k^s, f_k^t via Eq.6
end for

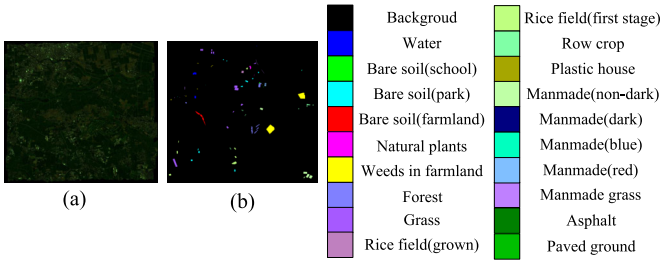


Fig. 6. Chikusei dataset. (a) False-color image. (b) Ground-truth map.

A. Datasets

In our experiments, we used six publicly available HSI datasets, namely, the Chikusei, Indian Pines, Salinas, HoustonU 2013, YanCheng, and XuZhou datasets, to assess the performance of the proposed method. Chikusei served as the source domain dataset and the other five were used as target domain datasets. Among them, Chikusei and HoustonU 2013 contain the shared semantic category "water" and are considered datasets with shared semantic classes.

1) Source Data: Chikusei: The Chikusei dataset was captured in July 2014 using the Headwall Hyperspec-VNIR-C sensor over Chikusei, Japan. It comprises 128 spectral bands ranging from 343 to 1081 nm. The spatial size is 2517×2335 pixels, with a spatial resolution of 2.5 m. There are a total of 19 land cover classes. The categories of the dataset and the number of samples in each category are listed in Table I. A false-color image and ground-truth map of the Chikusei dataset are shown in Fig. 6.

TABLE I
CLASS NAMES AND SAMPLE SIZES FOR THE CHIKUSEI DATASET

Class	Name	Pixels
1	Water	2345
2	Bare soil (school)	2859
3	Bare soil (park)	236
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (grown)	13369
10	Rice field (First stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (non-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

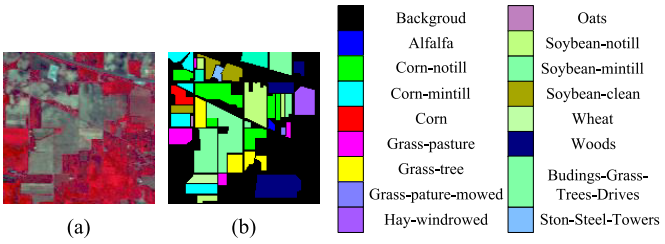


Fig. 7. Indian Pines dataset. (a) False-color image. (b) Ground-truth map.

2) Target Data: Indian Pines: The Indian Pines dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer AVIRIS sensor over the Indian Pines area in Indiana. The spatial size is 145×145 pixels, and it includes 224 spectral bands, 200 of which are effective. The dataset contains 16 different crop classes. The categories of the dataset and the number of samples in each category are listed in Table II. A false-color image and ground-truth map of the Indian Pines dataset are shown in Fig. 7.

Salinas: The Salinas dataset was also acquired by the AVIRIS sensor and covers the Salinas Valley in California. It consists of 224 spectral reflectance bands, 204 of which are effective. The spatial size is 512×217 pixels. The dataset includes 16 land cover classes. The categories of the dataset and the number of samples in each category are listed in Table II. A false-color image and ground-truth map of the Salinas dataset are shown in Fig. 8.

HoustonU 2013: The HoustonU 2013 dataset was obtained by the Compact Airborne Spectrographic Imager CASI-1500 sensor. It has a spatial size of 349×1905 pixels and includes 144 spectral bands ranging from 364 to 1046 nm. The land cover data are categorized into 15 classes. The categories of

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TABLE II
CLASS NAMES AND SAMPLE SIZES FOR DIFFERENT TARGET DATASETS

Class	Indian pines		Salinas		HoustonU 2013		XuZhou		YanCheng	
	Name	Pixels	Name	Pixels	Name	Pixels	Name	Pixels	Name	Pixels
1	Alfalfa	46	Brocoli_green_weeds_1	2009	Healthy grass	1251	Bareland_1	26396	Architecture	451
2	Corn-notill	1428	Brocoli_green_weeds_2	3726	Stressed grass	1254	Lakes	4027	River	214
3	Corn-mintill	830	Fallow	1976	Synthetic grass	697	Coals	2783	Reed	244
4	Corn	237	Fallow_rough_plow	1394	Trees	1244	Cement	5214	Paddy	3026
5	Grass-pasture	483	Fallow_smooth	2678	Soil	1242	Corp_1	13184	Fallow land	650
6	Grass-trees	730	Stubble	3979	Water	325	Trees	2436	Sea	2076
7	Grass-pasture-mowed	28	Celery	3579	Residential	1268	Bareland_2	6990	Offshore water	1558
8	Hey-windrowed	478	Grapes_untrained	11271	Commercial	1244	Crop_2	4777		
9	Oats	20	Soil_vinyard_develop	6203	Road	1252	Red_tiles	3070		
10	Soybean-notill	972	Corn_senesced_green_weeds	3278	Highway	1227				
11	Soybean-mintill	2455	Lettuce_romaine_4wk	1068	Railway	1235				
12	Soybean-clean	593	Lettuce_romaine_5wk	1927	Parking lot1	1233				
13	Wheat	205	Lettuce_romaine_6wk	916	Parking lot2	469				
14	Woods	1265	Lettuce_romaine_7wk	1070	Tennis court	428				
15	Buildings-Grass-Trees-Drives	386	Vinyard_untrained	7268	Running track	660				
16	Stone-Steel-Towers	93	Vinyard_vertical_trellis	1807						
	Total	10249		54129		15029		68877		8219

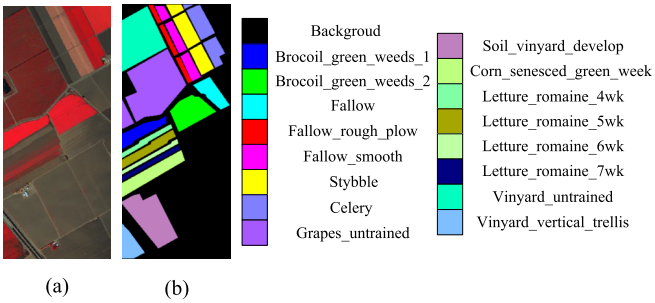


Fig. 8. Salinas dataset. (a) False-color image. (b) Ground-truth map.

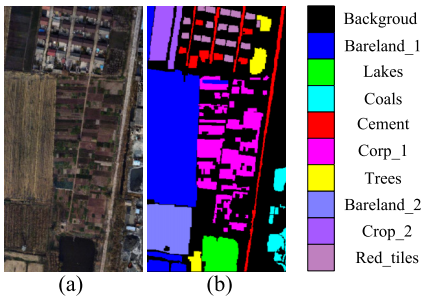


Fig. 10. XuZhou dataset. (a) False-color image. (b) Ground-truth map.

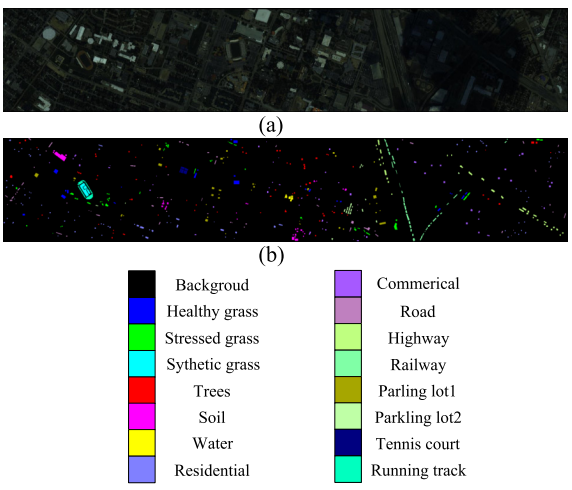


Fig. 9. HoustonU 2013 dataset. (a) False-color image. (b) Ground-truth map.

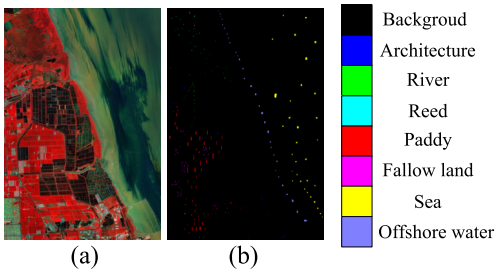


Fig. 11. YanCheng dataset. (a) False-color image. (b) Ground-truth map.

the dataset and the number of samples in each category are listed in Table II. A false-color image and ground-truth map of the HoustonU 2013 dataset are shown in Fig. 9.

XuZhou: The XuZhou dataset was acquired by an airborne HYSPEX hyperspectral camera over a peri-urban site in XuZhou. The dataset comprises 500 × 260 pixels, with 436 spectral bands retained after removing noisy bands in the range of 415–2508 nm. The scene is characterized by nine categories, as detailed in Table II. A false-color image and ground-truth map of the XuZhou dataset are shown in Fig. 10.

YanCheng: The YanCheng dataset was captured by the ZY1-02D satellite, equipped with a visible and near-infrared camera and an advanced hyperspectral imager (AHSI) sensor. The dataset has a spatial size of 1398 × 942 pixels and includes 147 spectral bands. The land cover data are categorized into seven classes. The categories of the dataset and the number of samples in each category are listed in Table II. A false-color image and ground-truth map of the YanCheng dataset are shown in Fig. 11.

B. Experimental Settings

For the source domain data, categories with fewer than 200 labeled samples were first removed. Then, 200 labeled samples from each remaining category were selected to form

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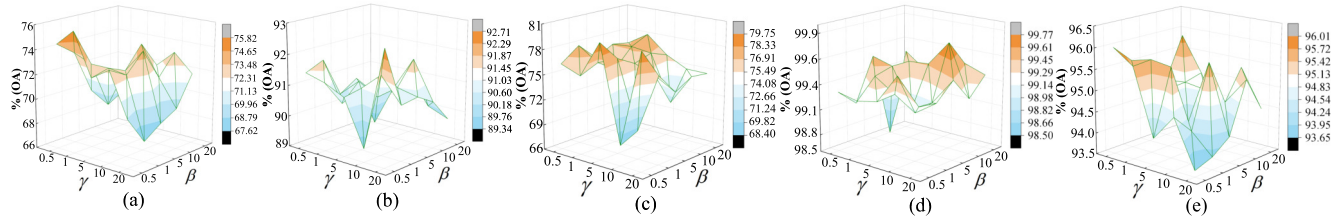


Fig. 12. Parameter analysis of β and γ for the proposed loss function. (a) Indian Pines, (b) Salinas, (c) HoustonU 2013, (d) YanCheng, and (e) XuZhou.

the training set. For the target domain, five labeled samples were randomly selected from each category, and Gaussian noise was used to augment these samples to a sample size of 200, forming the training set. The remaining samples were used as the test set. The size of the input spatial patch $w \times w$ was set to 9×9, and the number of spectral dimensions d for all datasets was reduced to 100 by the mapping module. The output dimensions of the feature extractor dim were set to 128 for the Salinas, HoustonU 2013, YanCheng, and XuZhou datasets and 64 for the Indian Pines dataset. For each episode, the number of support samples K was set to 1, the number of query samples Q was set to 19, and the number of categories N was set to the number of test categories in the target dataset. The number of episode iterations was set to 10000, the learning rate was set to 0.001, and Adam was used as the optimizer. The length of the domain-mixed dictionary DL was set to 100. The momentum coefficient b for the momentum encoder was set to 0.9 and the temperature coefficient for contrastive learning was set to 0.5.

C. Parameter Analysis

To validate the effects of different hyperparameters on model performance, we conducted analysis experiments on the intradomain category-level contrastive loss coefficient β and interdomain category-distribution-level alignment loss coefficient γ across five datasets. The values of β and γ were selected in the range of $\{0.5, 1, 5, 10, 20\}$. The trend of the classification overall accuracy (OA) for CDMLC is shown in Fig. 12.

For the Indian Pines dataset, the optimal parameter configuration is $\beta = 0.5$ and $\gamma = 1$, achieving the highest accuracy of 75.81%. When β is set to 0.5 or 1, and γ varies from 0.5 to 10, as well as along the diagonal line where β equals γ, the accuracy consistently exceeds 70%, indicating that the method demonstrates strong robustness to parameter selection on the Indian Pines dataset.

For the Salinas dataset, the optimal parameters are $\beta = 0.5$ and $\gamma = 20$, achieving the highest accuracy of 92.71%. Similar to the Indian Pines dataset, when β is set to 0.5 or 1 and γ ranges from 0.5 to 20, the overall performance is better than in other parameter ranges. Moreover, from a global perspective, the accuracy on this dataset exhibits relatively small fluctuations.

For the HoustonU 2013 dataset, the optimal parameters are $\beta = 0.5$ and $\gamma = 5$ or 20, resulting in the highest accuracy of 79.75%. Overall, the accuracy under larger γ values ($\gamma = 10$ and 20) is generally lower than that under smaller γ values ($\gamma = 0.5$, 1, and 5), indicating that excessively large γ may negatively affect performance.

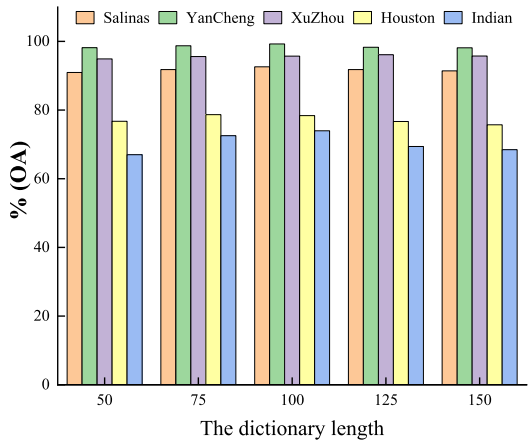


Fig. 13. Parameter analysis for the length of the domain-mixed dictionary.

For the YanCheng dataset, the optimal parameters are $\beta = 10$ and $\gamma = 10$, achieving the highest accuracy of 99.77%. Performance across all parameter combinations ranges between 98.5% and 99.77%, highlighting the stability and minimal sensitivity to parameter variations.

For the XuZhou dataset, the optimal parameters are $\beta = 0.5$ and $\gamma = 0.5$, achieving the highest accuracy of 96.01%. Relatively high accuracy is mainly observed when β is set to 0.5, 1, or around 20, while the lowest accuracy occurs when γ is set to 20.

In addition, we conducted an analysis experiment on the length of the domain-mixed dictionary. The dictionary length was set to $\{50, 75, 100, 125, 150\}$. We conducted the experiments on five target datasets, and the results are shown in Fig. 13. With increasing dictionary length, the overall classification accuracy first increased and then decreased. For the Salinas, YanCheng, and Indian Pines datasets, the optimal results were obtained when the dictionary length was set to 100. For the HoustonU 2013 dataset, the optimal results were achieved when the dictionary length was set to 75, and the suboptimal results were obtained when the dictionary length was set to 100. For the XuZhou dataset, the optimal results were achieved when the dictionary length was set to 125. Considering the results comprehensively, we chose 100 as the dictionary length in our experiments.

D. Comparison With State-of-the-Art Methods

1) Comparison Methods: To demonstrate the superiority of our CDMLC method, we compared it with two traditional CNN-based methods, three Transformer-based methods,

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TABLE III

CLASSIFICATION PERFORMANCE (IN PERCENT ± STB.DEV.) FOR THE INDIAN PINES DATASET

class	FDSSC [53]	DBMA [54]	SSFTT [55]	SpectralFormer [56]	GraphGST [57]	DFSL [16]	CMFSL [58]	DCFSL [42]	STBDIP [45]	Gia-CFSL [23]	ADAFSL [43]	CDMLC
1	34.98±19.52	20.21±5.11	93.66±8.01	91.71±14.59	99.39±1.06	90.24±8.68	96.59±4.78	96.75±4.60	98.54±1.95	90.24±12.91	93.17±7.14	98.78±2.11
2	81.59±8.97	52.99±11.28	44.20±7.22	47.10±10.12	49.63±6.32	45.21±14.50	44.96±11.42	35.37±8.24	34.01±9.25	45.08±9.65	55.60±10.06	49.83±3.85
3	35.95±31.31	38.96±4.14	55.59±4.91	35.98±15.40	64.27±10.74	47.19±3.64	55.27±9.42	52.20±5.32	56.07±12.25	48.06±5.24	63.47±4.67	64.18±6.24
4	50.69±4.37	34.57±10.11	78.79±11.18	43.19±2.93	75.00±6.77	75.29±9.58	85.95±6.77	83.76±4.88	89.31±7.64	80.78±8.32	84.74±7.36	84.36±14.66
5	89.02±10.94	52.38±18.23	83.47±6.58	73.14±7.81	75.00±12.38	79.50±3.14	79.62±6.97	76.43±2.56	77.87±7.44	73.14±6.90	79.58±5.48	84.36±4.48
6	87.21±12.69	84.41±8.30	92.11±2.09	96.72±2.59	94.59±3.01	76.78±1.47	80.06±10.54	93.61±3.38	93.88±1.19	80.14±10.36	89.05±4.85	92.04±4.39
7	18.20±9.91	17.85±4.04	100.00±0.00	99.13±1.95	100.00±0.00	100.00±0.00	99.13±1.74	100.00±0.00	100.00±0.00	99.57±1.30	99.13±1.74	100.00±0.00
8	88.89±3.70	98.41±1.62	79.79±8.20	95.43±2.32	84.73±19.54	89.64±1.38	90.57±7.22	93.23±3.96	91.21±7.09	90.97±7.12	91.16±3.70	97.99±1.06
9	3.26±1.00	2.29±0.84	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	97.33±5.33	98.00±3.06	100.00±0.00	100.00±0.00
10	62.63±9.21	61.81±9.28	66.58±7.20	62.81±9.84	75.62±13.84	57.08±6.00	65.89±4.67	63.91±3.74	70.92±7.60	62.97±10.76	69.68±5.83	60.81±8.13
11	79.26±10.90	74.27±2.84	51.99±12.89	58.12±8.03	44.15±5.05	54.54±16.36	55.87±16.06	66.33±0.37	66.52±9.47	60.49±11.40	54.55±18.25	71.15±8.47
12	54.91±26.45	42.26±11.27	41.84±13.38	28.10±6.92	69.09±11.09	41.10±13.82	43.64±13.63	41.21±1.95	47.65±12.80	47.28±11.12	53.33±18.70	48.47±16.17
13	38.21±7.12	54.51±10.59	98.90±0.96	97.70±2.61	98.88±0.65	99.50±0.71	96.20±6.42	99.00±1.08	100.00±0.00	96.65±5.24	98.60±1.62	99.88±0.22
14	95.86±2.86	96.28±5.63	90.03±7.65	80.75±5.94	81.77±10.82	75.50±3.89	87.16±5.19	87.30±1.80	86.86±8.96	78.94±8.96	82.14±10.61	90.69±5.03
15	57.53±24.90	53.40±15.27	68.34±5.95	55.07±6.94	80.12±13.22	56.61±9.85	77.38±10.95	57.13±10.20	84.46±7.77	63.91±10.70	80.05±8.56	77.10±6.63
16	22.85±6.48	44.26±6.14	100.00±0.00	97.05±4.06	100.00±0.00	100.00±0.00	99.55±0.91	95.45±4.04	100.00±0.00	99.09±1.22	97.95±2.53	99.72±0.49
OA	57.62±8.73	55.90±4.46	65.25±2.97	62.44±2.61	66.62±2.42	61.24±2.24	65.86±2.59	66.63±1.61	68.99±2.72	64.40±3.03	68.76±3.40	**72.52±1.07**
AA	56.31±3.41	51.80±1.24	77.84±1.00	72.62±2.29	80.77±1.24	74.26±1.07	78.61±1.22	77.61±1.73	80.91±1.86	75.96±2.25	80.76±0.79	**82.12±0.62**
kappa	53.72±9.07	50.91±4.56	61.10±3.27	57.52±2.78	62.89±2.52	56.43±2.23	61.63±2.79	62.27±1.79	65.02±3.02	59.85±3.34	64.97±3.44	**68.79±0.92**

TABLE IV

CLASSIFICATION PERFORMANCE (IN PERCENT ± STB.DEV.) FOR THE SALINAS DATASET

class	FDSSC [53]	DBMA [54]	SSFTT [55]	SpectralFormer [56]	GraphGST [57]	DFSL [16]	CMFSL [58]	DCFSL [42]	STBDIP [45]	Gia-CFSL [23]	ADAFSL [43]	CDMLC
1	94.12±11.77	81.32±22.97	99.75±0.15	97.31±2.36	99.83±0.25	99.58±0.55	99.95±0.08	99.79±0.14	99.50±1.00	98.27±1.29	98.99±1.03	99.84±0.14
2	99.15±1.70	98.37±2.74	99.56±0.50	99.88±0.21	99.98±0.02	99.61±0.40	99.52±0.88	99.73±0.37	99.96±0.09	99.68±0.34	99.87±0.18	100.00±0.00
3	96.03±2.96	91.45±7.47	99.76±0.25	98.64±2.33	78.74±5.76	82.89±11.50	96.64±1.75	96.50±1.13	94.76±8.80	88.86±10.42	98.61±0.85	90.98±5.51
4	91.10±3.17	85.17±6.84	94.51±11.87	96.85±3.44	99.44±0.24	97.36±2.33	99.08±1.31	99.34±0.53	98.52±1.71	99.32±0.47	99.52±0.39	99.89±0.11
5	99.36±0.80	95.67±7.29	95.78±2.22	96.18±3.02	96.82±2.95	91.89±4.13	92.86±1.83	91.42±1.23	91.30±3.63	89.05±5.61	95.91±2.04	91.02±2.91
6	99.90±0.19	99.82±0.35	99.72±0.46	99.79±0.37	99.92±0.07	98.11±2.58	98.65±1.75	99.93±0.09	99.95±0.06	98.56±1.61	99.97±0.00	99.22±1.36
7	99.30±1.27	99.46±0.63	99.80±0.18	99.01±1.90	97.15±0.52	99.29±0.51	99.74±0.32	99.31±0.42	99.82±0.13	99.12±1.34	99.02±0.70	99.82±0.19
8	79.35±11.10	75.42±8.74	74.82±13.15	73.24±9.27	71.55±7.18	75.40±4.95	78.75±3.49	75.15±5.35	83.18±5.05	75.99±12.94	80.03±1.93	81.38±3.00
9	99.12±0.74	97.60±2.33	97.76±4.89	99.88±0.16	99.51±0.23	99.14±0.95	98.75±1.74	98.55±1.31	98.85±1.39	97.91±3.14	99.39±0.87	99.49±0.88
10	89.53±4.16	79.82±7.37	92.46±1.60	89.50±6.02	82.08±12.43	72.52±13.49	90.26±2.62	87.51±4.18	87.75±2.88	86.06±7.31	90.99±1.44	89.28±3.76
11	89.93±6.94	68.59±12.50	96.86±3.31	99.63±0.48	95.72±1.87	95.80±2.32	99.25±0.56	98.87±1.42	97.87±1.56	97.29±3.37	98.68±0.55	99.32±0.66
12	98.58±1.69	91.73±12.49	99.86±0.21	98.69±1.05	96.96±2.69	97.55±2.44	99.81±0.30	99.94±0.12	99.53±0.51	98.51±1.79	99.91±0.12	99.57±0.45
13	99.50±0.53	78.02±6.21	89.73±8.44	95.96±5.19	98.63±0.58	97.69±2.33	99.25±0.25	99.17±0.60	99.39±0.97	98.49±1.09	99.71±0.21	98.14±1.63
14	92.12±4.35	87.07±8.00	97.32±1.58	93.75±7.38	97.21±1.25	99.78±0.09	98.70±0.69	98.54±1.10	97.69±0.82	98.09±1.45	99.78±0.31	99.48±0.34
15	66.07±16.76	62.69±9.83	74.43±6.08	80.41±10.44	91.70±5.10	77.14±7.28	72.50±9.23	79.70±5.54	70.99±10.17	74.80±4.93	73.95±7.24	77.83±8.07
16	99.15±1.31	95.66±2.13	96.08±2.38	95.53±5.09	93.52±7.61	90.73±3.40	94.26±4.16	94.89±3.89	92.70±3.99	88.29±8.25	93.04±4.45	94.41±4.33
OA	86.89±3.40	83.57±1.70	89.75±1.99	90.24±1.62	90.18±0.93	88.20±1.95	90.24±0.84	90.29±0.90	90.67±1.39	88.73±2.05	91.07±1.40	**91.32±1.01**
AA	93.27±1.80	86.74±1.05	94.26±1.18	94.64±1.28	93.68±1.05	92.16±1.73	94.87±0.52	94.90±0.27	94.48±0.82	93.01±1.51	**95.46±0.79**	94.98±0.78
kappa	85.43±3.80	81.75±1.90	88.61±2.17	89.17±1.79	89.10±1.03	86.90±2.14	89.16±0.95	89.22±0.98	89.61±1.56	87.49±2.86	90.07±1.57	**90.36±1.12**

one FSL method based on meta-learning, and five cross-domain FSL methods. The CNN-based methods included the fast dense spectral–spatial convolution network (FDSSC) [53] and the double-branch multiattention network (DBMA) [54]. FDSSC combines the advantages of SSRN [59] and DenseNet [60] and separately learns spectral and spatial features via a dense connection structure. DBMA uses two parallel branch networks (spatial branch and spectral branch) to extract spatial and spectral features separately while introducing multiple attention modules for each branch. The Transformer-based methods included the GraphGST [57], SpectralFormer [56], and SSFTT [55]. The FSL method based on meta-learning is the DFSL method [16]. This method trains on the source domain and tests on the target domain without considering the data distribution differences between the source and target domains. On the basis of DFSL method, the five cross-domain FSL methods all consider the domain gap issue and perform domain alignment; these five methods were DCFSL [42], class covariance metric FSL (CMFSL) [58], multilevel relation learning (STBDIP) [45], Gia-CFSL [23], and ADAFSL [43]. CMFSL uses class covariance metrics to improve the decision boundaries of nonlinear classifiers.

For all the methods mentioned above and the proposed method, five labeled samples per class were randomly selected as training samples, and the parameters were set to the default values specified in their open-source code or original papers. For the cross-domain FSL methods and FSL methods, the Chikusei dataset was used as the source domain. Considering that the small number of training samples could significantly impact model performance, we repeated each experiment five times and report the average results and standard deviation to increase the reliability of the experimental results.

2) Results and Analysis: We use OA, average accuracy (AA), and the kappa coefficient to evaluate model performance. The experimental results are shown in Tables III–VII. The following observations can be made from the data presented in Tables III–VII.

Our method achieved state-of-the-art performance on all five benchmark datasets. On the Indian Pines dataset, CDMLC outperformed the next-best method (ADAFSL) with significant improvements of 3.72% in OA, 1.36% in AA, and 3.82% in kappa. For the Salinas dataset, CDMLC achieved a 0.25% higher OA and 0.29% higher kappa, albeit with a marginal 0.48% decrease in AA, compared to ADAFSL. On the HoustonU 2013 dataset, CDMLC demonstrated superior performance over CMFSL, with gains of 1.14% in OA, 0.3% in AA, and 1.21% in kappa. For the XuZhou dataset, CDMLC surpassed CMFSL by 0.22% in OA, 0.36% in AA, and 0.13% in kappa. Finally, on the YanCheng dataset, CDMLC achieved a 0.12% higher OA and 0.28% higher kappa compared to STBDIP. These consistent improvements across all datasets demonstrate that our method is effective regardless of whether the source and target domains share semantic categories, thereby confirming its robustness and adaptability to various domain scenarios.

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TABLE V

CLASSIFICATION PERFORMANCE (IN PERCENT ± STB.DEV.) FOR THE HOUSTONU 2013 DATASET

class	FDSSC [53]	DBMA [54]	SSFTT [55]	SpectralFormer [56]	GraphGST [57]	DFSL [16]	CMFSL [58]	DCFSL [42]	STBDIP [45]	Gia-CFSL [23]	ADAFSL [43]	CDMLC
1	64.59±16.87	65.05±5.85	83.40±6.80	87.71±8.56	89.43±10.22	87.08±8.45	84.72±0.76	89.89±6.10	90.69±7.28	90.91±6.80	94.28±6.91	91.45±6.80
2	72.25±20.27	79.65±6.08	82.03±3.38	77.04±9.28	84.95±7.98	79.85±10.76	78.27±15.53	78.99±12.24	81.33±13.68	80.93±6.83	75.98±13.20	84.63±5.46
3	77.16±22.89	93.39±11.27	99.37±0.65	98.03±3.36	88.95±7.39	94.03±4.40	96.65±1.59	96.13±1.23	98.06±0.85	94.25±4.61	98.31±1.49	97.57±2.44
4	48.63±17.25	73.7±11.28	87.34±3.54	90.62±7.48	82.71±13.10	92.06±1.16	93.69±2.17	86.55±9.72	94.71±2.46	91.11±2.03	88.30±3.93	93.16±2.95
5	83.07±11.61	91.2±7.07	95.96±9.04	95.71±8.64	99.76±0.42	91.65±7.03	96.49±6.13	95.75±2.54	90.14±11.33	94.08±4.51	95.80±3.01	96.49±3.19
6	94.15±6.52	84.46±15.29	85.31±10.37	92.19±5.79	91.02±9.72	75.94±7.30	83.56±7.33	79.75±4.93	82.62±4.96	82.06±7.54	76.04±4.71	84.94±5.60
7	58.99±12.16	66.3±3.44	53.35±19.49	58.84±10.11	62.53±3.90	68.73±6.12	71.62±9.25	71.04±8.58	70.06±17.88	66.44±13.13	53.71±9.78	66.63±12.25
8	76.06±17.95	73.17±17.75	39.37±7.70	57.97±5.70	44.35±6.03	35.81±6.91	49.65±4.64	37.90±7.27	41.00±7.92	44.25±10.43	42.02±7.52	49.81±10.03
9	58.00±12.50	68.42±11.86	59.12±18.59	52.67±5.55	66.44±10.72	53.76±9.61	67.54±9.41	64.72±15.09	68.77±12.99	62.47±13.26	64.29±13.89	67.46±9.98
10	72.77±16.46	63.26±14.03	71.33±4.36	74.98±11.94	56.14±9.78	73.95±11.75	65.66±14.80	65.55±10.35	57.05±12.26	75.16±10.61	69.67±16.92	70.98±11.52
11	67.01±27.96	59.02±13.98	76.08±16.91	78.03±7.72	58.60±12.45	56.48±10.98	60.46±8.84	61.14±6.36	59.51±11.50	61.93±10.54	55.88±5.73	63.46±3.37
12	75.40±14.67	82.20±6.61	62.49±11.06	68.65±23.28	64.60±6.91	66.40±12.59	69.77±13.52	75.33±5.29	67.67±3.33	73.57±13.49	67.73±9.53	73.70±11.41
13	32.11±5.15	63.24±17.01	93.23±3.03	42.76±11.16	81.73±5.63	89.15±5.06	95.00±1.30	93.23±1.66	91.68±5.04	88.58±2.92	48.28±8.25	82.37±5.28
14	83.86±11.89	90.49±11.30	95.70±8.83	82.41±12.19	98.35±1.15	92.20±5.20	95.65±7.90	89.41±6.06	89.08±2.61	94.42±1.38	82.11±4.63	93.57±6.91
15	89.96±7.88	90.22±4.74	99.91±0.14	100.00±0.00	100.00±0.00	94.25±4.02	98.56±1.68	97.89±0.88	98.14±2.38	92.09±5.20	92.01±9.70	95.63±3.82
OA	61.68±5.47	72.23±5.01	75.29±2.10	76.06±2.71	74.62±2.43	74.01±0.53	77.43±1.60	76.15±2.34	75.78±3.06	77.00±1.02	72.70±1.60	**78.57±1.13**
AA	70.27±2.18	76.25±3.25	78.93±1.79	77.17±2.84	77.97±2.25	76.75±1.14	80.49±1.31	78.88±2.01	78.70±2.63	79.48±0.84	73.63±1.91	**80.79±0.96**
kappa	58.73±5.90	70.02±5.39	73.32±2.25	74.10±2.94	72.61±2.62	71.92±0.59	75.63±1.72	74.23±2.53	73.86±3.29	75.15±1.10	70.48±1.73	**76.84±1.21**

TABLE VI

CLASSIFICATION PERFORMANCE (IN PERCENT ± STB.DEV.) FOR THE XUZHOU DATASET

class	FDSSC [53]	DBMA [54]	SSFTT [55]	SpectralFormer [56]	GraphGST [57]	DFSL [16]	CMFSL [58]	DCFSL [42]	STBDIP [45]	Gia-CFSL [23]	ADAFSL [43]	CDMLC
1	99.36±0.52	98.79±6.09	87.94±1.52	87.43±1.83	87.25±2.92	95.31±0.82	94.47±2.87	94.04±1.09	92.37±4.08	92.52±2.29	91.54±2.16	93.40±1.60
2	99.32±0.99	98.23±2.55	97.69±0.79	97.41±1.02	99.36±0.08	97.24±1.16	97.71±0.92	96.31±0.55	98.94±0.32	96.69±1.84	97.53±0.92	97.79±1.16
3	79.80±12.61	75.92±12.48	92.54±1.43	90.67±8.15	97.76±0.96	87.37±4.63	93.53±5.34	93.30±2.11	96.97±1.69	86.56±4.31	94.81±1.83	95.19±3.16
4	95.81±2.01	94.30±3.04	75.81±5.91	78.69±8.14	87.69±8.09	89.86±3.59	93.32±4.66	82.20±3.17	88.57±9.71	88.82±2.43	87.17±4.74	91.16±4.93
5	97.04±0.16	95.65±1.12	86.71±6.46	80.05±9.05	95.51±2.18	87.42±3.47	90.07±8.10	82.46±5.51	96.07±3.28	93.04±4.04	94.11±2.00	95.28±3.32
6	79.80±10.22	74.78±9.58	96.57±2.04	87.11±6.87	90.82±8.56	87.39±2.14	96.43±2.44	90.00±8.17	97.30±1.63	87.84±6.35	93.68±5.35	89.15±7.47
7	78.59±5.13	78.67±5.54	89.42±8.97	76.53±15.67	99.03±0.46	95.10±2.36	95.89±3.58	99.03±0.15	96.91±2.32	97.84±1.07	95.77±2.40	97.53±2.03
8	88.02±9.27	80.97±13.20	98.2±0.81	98.74±0.93	96.00±6.46	99.49±0.54	98.54±0.91	99.47±0.29	97.98±2.60	99.70±0.11	99.66±0.36	99.17±0.87
9	85.55±13.13	96.45±3.13	98.23±1.42	97.16±0.81	95.14±1.52	96.31±1.19	95.07±4.24	88.56±3.07	97.77±0.99	95.15±5.06	97.31±2.97	96.67±3.23
OA	92.18±1.56	91.04±0.93	89.17±0.81	86.17±2.89	92.28±0.55	93.21±1.09	94.21±1.23	91.53±0.74	94.62±1.99	93.33±1.04	93.51±1.21	**94.74±0.78**
AA	89.28±1.98	88.20±1.15	91.46±1.09	88.20±2.68	94.28±1.29	92.83±0.77	95.00±1.14	91.71±0.94	**95.87±1.50**	93.13±0.73	94.62±1.08	95.04±0.83
kappa	90.24±1.91	88.83±1.17	86.52±1.01	82.85±3.50	90.35±0.66	91.44±1.35	92.72±1.54	89.36±0.92	93.25±2.46	91.62±1.28	91.87±1.47	**93.38±0.96**

TABLE VII

CLASSIFICATION PERFORMANCE (IN PERCENT ± STB.DEV.) FOR THE YANCHENG DATASET

class	FDSSC [53]	DBMA [54]	SSFTT [55]	SpectralFormer [56]	GraphGST [57]	DFSL [16]	CMFSL [58]	DCFSL [42]	STBDIP [45]	Gia-CFSL [23]	ADAFSL [43]	CDMLC
1	99.46±1.09	100.00±0.00	72.42±15.43	67.6±17.88	60.20±15.08	91.48±5.28	90.67±10.38	87.62±9.31	89.52±11.61	88.79±10.45	83.18±8.25	95.44±0.69
2	78.31±11.77	89.12±8.77	99.88±0.24	93.3±7.29	88.88±8.43	99.28±0.72	99.52±0.61	98.18±3.40	98.56±0.90	100.00±0.00	95.81±5.63	95.69±4.23
3	100.00±0.00	88.49±6.19	94.77±2.61	99.06±1.88	97.49±2.23	100.00±0.00	99.67±0.67	99.41±1.17	97.18±3.09	99.50±1.00	99.37±1.09	100.00±0.00
4	99.53±0.48	99.33±0.47	94.99±2.50	99.48±0.37	99.64±0.33	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	99.98±0.03	100.00±0.00
5	99.66±0.42	99.79±0.28	95.74±2.64	97.09±2.22	88.18±5.90	94.53±0.13	96.65±2.73	95.29±2.41	100.00±0.00	94.54±2.55	95.81±2.42	98.40±2.27
6	100.00±0.00	100.00±0.00	100.00±0.00	99.41±0.68	97.57±2.60	97.25±0.16	99.98±0.02	99.03±1.18	99.98±0.02	97.01±5.96	98.58±1.41	100.00±0.00
7	94.34±11.33	95.89±8.21	98.6±2.80	98.84±2.32	96.28±1.28	100.00±0.00	99.70±0.53	99.25±1.04	99.66±0.59	98.87±1.65	98.58±1.41	99.16±0.64
OA	97.50±2.97	98.03±1.83	95.89±1.44	97.24±1.21	95.09±1.31	98.39±0.33	99.14±0.77	98.50±0.74	98.97±0.74	97.97±1.60	98.13±0.71	**99.36±0.22**
AA	95.90±2.20	96.09±1.53	93.77±2.26	93.54±3.21	89.75±3.06	97.51±0.82	98.03±1.73	96.97±1.28	97.36±1.77	96.96±1.85	96.01±1.03	**98.39±0.58**
kappa	96.69±3.92	97.38±2.43	94.57±1.90	96.34±1.61	93.48±1.73	97.86±0.44	98.86±0.91	98.01±0.99	98.64±0.98	97.31±2.11	97.52±0.94	**99.14±0.29**

TABLE VIII

EVALUATIONS OF THE CONTRIBUTIONS OF VARIOUS MODULES ON DIFFERENT TARGET DATASETS

L_{fsl}	$L_{intra}^{\text{non-mixed}}$	L_{intra}	L_{μ}	L_{σ}	Indian Pines		Salinas		HoustonU 2013		YanCheng		XuZhou	
					OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa
✓					65.36±0.90	60.59±1.13	88.30±0.70	87.01±0.76	73.67±1.83	71.57±1.99	98.92±0.65	98.57±0.86	93.21±1.09	91.44±1.35
✓	✓				-	-	-	-	75.99±0.17	74.07±0.18	-	-	-	-
✓		✓			67.19±3.14	62.97±3.45	89.83±1.06	88.70±1.16	76.78±1.00	74.92±1.07	98.99±0.61	98.65±0.80	93.51±0.77	91.84±0.95
✓		✓	✓		70.44±3.89	66.91±4.19	89.93±2.30	88.80±2.56	77.76±1.18	75.98±1.27	99.17±0.12	98.90±0.17	94.11±0.56	92.59±0.69
✓		✓	✓	✓	72.52±1.07	68.79±0.92	91.32±1.01	90.36±1.12	78.57±1.13	76.84±1.21	99.36±0.22	99.14±0.29	94.74±0.78	93.38±0.96

Meanwhile, we can observe that in the Indian Pines dataset, the classes "corn-notill" and "corn-mintill," which represent different tillage practices of the same crop (corn), exhibit similar spectral characteristics. This similarity makes them difficult to distinguish and results in lower accuracy. In the Salinas dataset, the class "Brocoli_green_weeds_2" achieves 100% accuracy with a standard deviation of 0.00. This class of vegetation has low reflectance in the visible light band and high reflectance in the near-infrared band, making its features distinct and easy for the model to differentiate. On the other hand, the spectral characteristics of the "Vinyard_untrained" class are easily influenced by canopy structure, leading to substantial fluctuations in accuracy. In the HoustonU 2013 dataset, compared to other cross-domain methods, the "water" class achieves the highest accuracy in our method, which demonstrates the necessity of considering semantic consistency between the source and target domains. For the YanCheng and XuZhou datasets, the high classification accuracy for each category demonstrates the excellent separability of the classes in these datasets.

Furthermore, FDSSC and DBMA exhibited the worst classification performance. Although these methods can extract discriminative features from HSIs, traditional CNN-based approaches are not efficient in addressing the few-shot problem. The classification performance of the DFSL method was better than that of the CNN-based methods. The results indicate that the meta-learning-based FSL methods can indeed learn classification models with strong generalization

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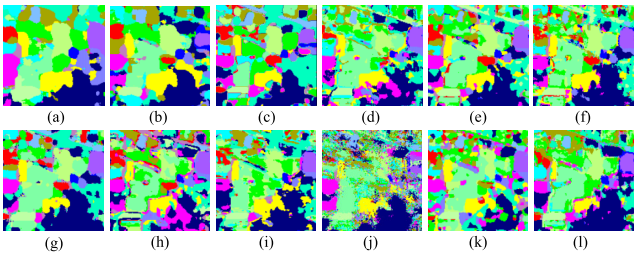


Fig. 14. Classification maps of the Indian Pines dataset. (a) FDSSC, (b) DBMA, (c) GraphGST, (d) DFSL, (e) CMFSL, (f) DCFSL, (g) STBDIP, (h) Gia-CFSL, (i) ADAFSL, (j) SpectralFormer, (k) SSFTT, and (l) CDMLC.

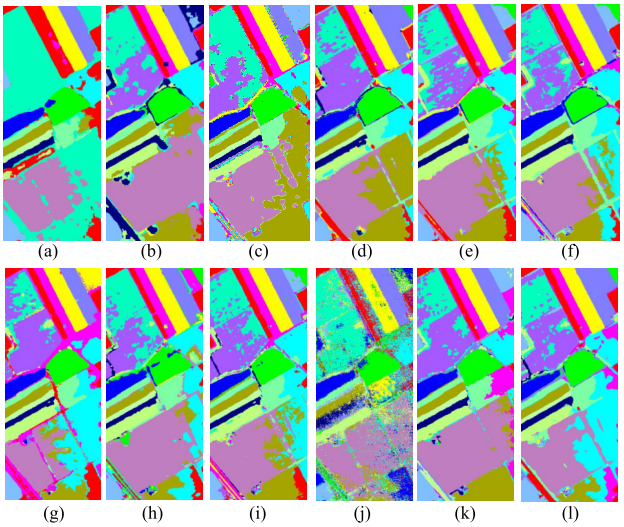


Fig. 15. Classification maps of the Salinas dataset. (a) FDSSC, (b) DBMA, (c) GraphGST, (d) DFSL, (e) CMFSL, (f) DCFSL, (g) STBDIP, (h) Gia-CFSL, (i) ADAFSL, (j) SpectralFormer, (k) SSFTT, and (l) CDMLC.

capabilities from the source dataset. However, Transformer-based methods like SpectralFormer and SSFTT have demonstrated competitive effectiveness in cross-domain FSL tasks. This shows the exceptional capability of Transformers in capturing discriminative features.

Finally, the methods incorporating domain alignment modules, such as the CMFSL, DCFSL, STBDIP, Gia-CFSL, and ADAFSL methods, outperformed the DFSL method. The results indicate that the domain gap is one of the main issues limiting the ability of meta-learning-based methods to truly solve the few-shot problem. Considering the distribution differences between different datasets is essential for enhancing the classification performance of cross-domain FSL methods.

3) Visual Comparisons: Figs. 14–18 visually illustrate the classification performance of the compared methods on five datasets. We can observe that the classification maps of FDSSC and DBMA are smoother, but they suffer from large-area misclassification, which is the main reason for their lower classification accuracy. In contrast, the classification maps of the other comparative methods show fewer large-area misclassifications. Compared with FDSSC and DBMA, although our method resulted in more noise, there is a noticeable reduction in large-area misclassification in the classification maps. Compared with both the FSL method and other cross-domain FSL methods, the classification maps of our method are smoother and have fewer misclassified areas.

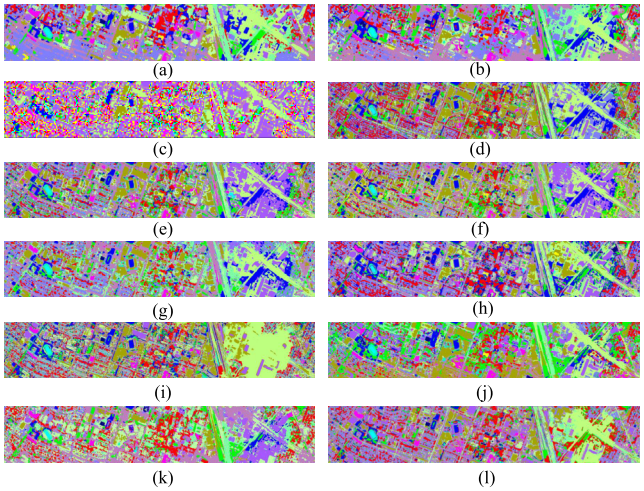


Fig. 16. Classification maps of the HoustonU 2013 dataset. (a) FDSSC, (b) DBMA, (c) GraphGST, (d) DFSL, (e) CMFSL, (f) DCFSL, (g) STBDIP, (h) Gia-CFSL, (i) ADAFSL, (j) SpectralFormer, (k) SSFTT, and (l) CDMLC.

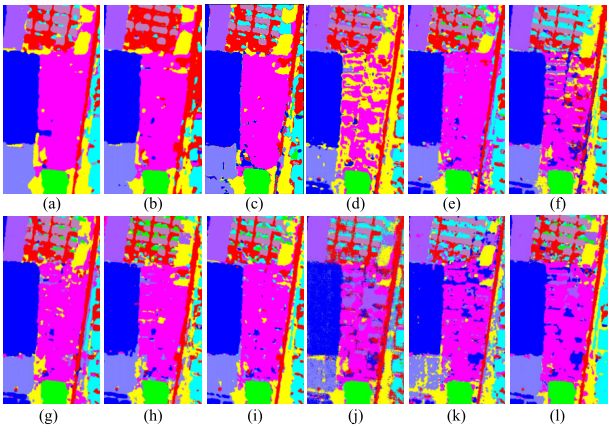


Fig. 17. Classification maps of the XuZhou dataset. (a) FDSSC, (b) DBMA, (c) GraphGST, (d) DFSL, (e) CMFSL, (f) DCFSL, (g) STBDIP, (h) Gia-CFSL, (i) ADAFSL, (j) SpectralFormer, (k) SSFTT, and (l) CDMLC.

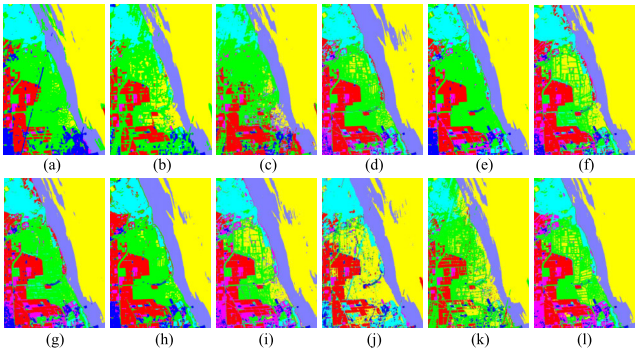


Fig. 18. Classification maps of the YanCheng dataset. (a) FDSSC, (b) DBMA, (c) GraphGST, (d) DFSL, (e) CMFSL, (f) DCFSL, (g) STBDIP, (h) Gia-CFSL, (i) ADAFSL, (j) SpectralFormer, (k) SSFTT, and (l) CDMLC.

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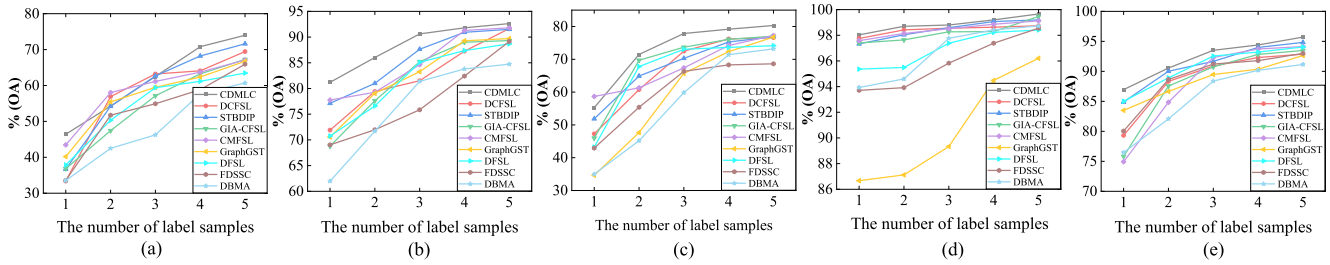


Fig. 19. Classification accuracy (%) with different numbers of labeled samples per class. (a) Indian Pines, (b) Salinas, (c) HoustonU 2013, (d) YanCheng, and (e) XuZhou.

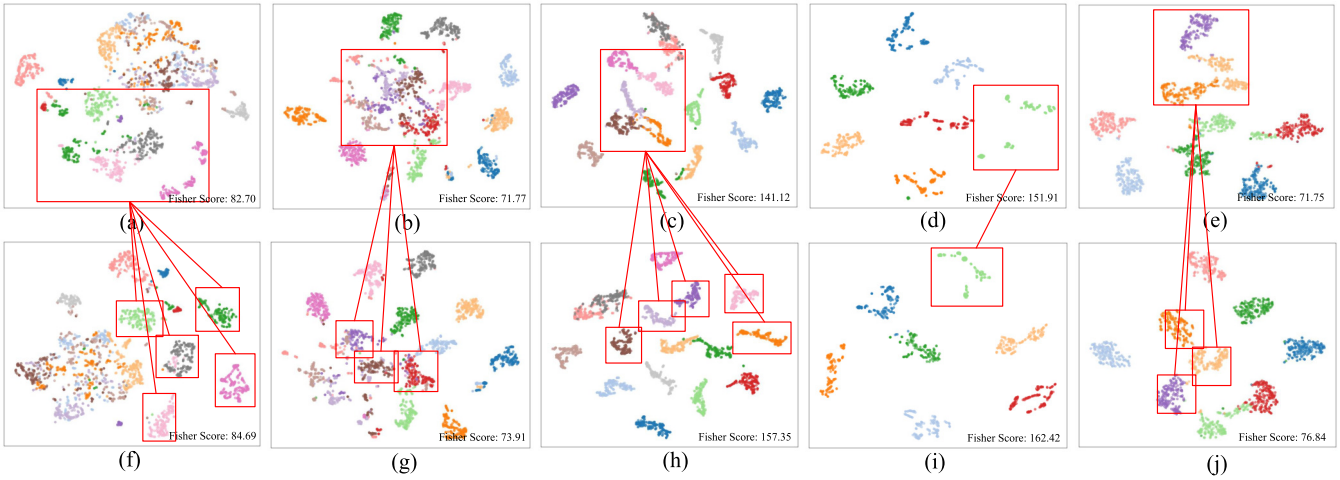


Fig. 20. (a) FSL distribution for the Indian Pines. (b) FSL distribution for the HoustonU 2013. (c) FSL distribution for the Salinas. (d) FSL distribution for the YanCheng. (e) FSL distribution for the XuZhou. (f) CDMLC distribution for the Indian Pines. (g) CDMLC distribution for the HoustonU 2013. (h) CDMLC distribution for the Salinas. (i) CDMLC distribution for the YanCheng. (j) CDMLC distribution for the XuZhou.

E. Stability Analysis

To validate the impact of the number of labeled samples used for training in the target domain on model performance, we conducted experiments by randomly selecting 1–5 labeled samples per category. The classification results of different methods with varying numbers of labeled samples are shown in Fig. 19. We can see that as the number of labeled samples in the target domain increased, the OA metric for each method also increased. Additionally, our method consistently achieved the best results on the Salinas, YanCheng, and XuZhou datasets. On the HoustonU 2013 dataset, our method achieved the best results under all conditions except when there was only one labeled sample. On the Indian Pines dataset, our method achieved optimal performance when the number of labeled samples was 1, 4, or 5.

F. Ablation Experiments

To validate the effectiveness of each module proposed in this article, we conducted ablation experiments on the loss of each core module, which included mainly: 1) instance-level FSL loss L_{fsl}; 2) intradomain contrastive loss constructed with nondomain-mixed dictionaries $L_{intra}^{non-mixed}$; 3) intradomain contrastive loss constructed with domain-mixed dictionaries L_{intra}; 4) mean distribution alignment loss from the inter-domain category-distribution-level alignment module L_{μ}; and 5) variance distribution alignment loss from the interdomain category-distribution-level alignment module L_{σ}.

As shown in Table VI, when the intradomain contrastive loss L_{intra} was introduced into the baseline meta-learning model (the third line), the OA increased by 1.83%, 1.53%, 3.11%, 0.07%, and 0.30% on the Indian Pines, Salinas, HoustonU 2013, YanCheng, and XuZhou datasets, respectively. In particular, the HoustonU 2013 dataset contains the same semantic categories as the source domain dataset; therefore, we tested the classification performance with and without the use of the domain-mixed dictionaries on the HoustonU 2013 dataset (the second line). The OA improved by 0.79% when the intradomain contrastive loss constructed with domain-mixed dictionaries was used compared with that with nondomain-mixed dictionaries. The results demonstrate the effectiveness of using domain-mixed dictionaries for the intradomain contrastive loss when the source and target domains contain the same semantic categories. Additionally, we tested the influence of the mean distribution and variance distribution on the performance of domain alignment. We noticed that after introducing the mean distribution alignment loss L_{μ} (the fourth line) and variance distribution alignment loss L_{σ} (the fifth line) in the category-distribution-level alignment module, the OAs on the Indian Pines, Salinas, HoustonU 2013, YanCheng, and XuZhou datasets increased to 72.52%, 91.32%, 78.57%, 99.36%, and 94.74%, respectively. The progressive

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improvements in accuracy with the addition of each module further validate the effectiveness of our method.

G. Feature Visualization

To demonstrate the effectiveness of our method in learning domain-invariant feature representations through multilevel contrastive learning, we conducted feature visualization using t-distributed stochastic neighbor embedding (t-SNE). Fig. 20 compares the embedded features processed by both the baseline FSL approach and our proposed CDMLC method across five benchmark datasets: Indian Pines, HoustonU 2013, Salinas, YanCheng, and XuZhou. For each class, we randomly selected 200 samples (or all available samples for classes with fewer than 200 instances). From the figure, we can observe that the features learned by FSL exhibit significant class overlap and ambiguous cluster boundaries. In contrast, the features learned by CDMLC demonstrate superior class separability and intraclass compactness, as evidenced by the formation of well-defined cluster structures with minimal interclass interference. This clear separation highlights the ability of CDMLC to preserve category structures and achieve effective domain alignment. In addition, we use the Fisher score [61] to quantitatively evaluate the separability of features, with the specific values annotated in the distribution maps. The results show that our method achieves consistent improvements across all datasets.

V. Conclusion and Future Work

This article proposes a novel method for addressing the cross-domain few-shot HSI classification task, named the CDMLC method. In this method, an intradomain category-level feature learning module and an interdomain category-distribution-level alignment module are constructed on the basis of the concept of multilevel contrastive learning. First, for the intradomain category-level feature learning module, we utilize the category priors of the training samples to construct a category-guided domain-mixed dictionary, which achieves compact intraclass feature representations for both domains. For the interdomain category-distribution-level alignment module, we use supervised GMM and unsupervised GMM to obtain category-level distributions for the source and target domains and then design a cross-domain category-distribution-level contrastive loss. This module achieves a category-level distribution alignment between the source and target domains while ensuring interclass separability. Extensive experiments and analyses on five widely used benchmark HSI datasets demonstrate that the proposed CDMLC method has strong generalization performance and excellent classification capabilities compared with current state-of-the-art methods.

While our method demonstrates strong performance in single-source domain adaptation, recent studies [62] have highlighted the potential of multisource domain to address large real-world domain shifts. In future work, we plan to extend our framework to incorporate multisource domains, leveraging their complementary information to further enhance generalization capacity in cross-scene remote sensing image classification.

References

[1] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] C. Shi, P. Han, M. Zhao, L. Fang, Q. Miao, and C.-M. Pun, "Adaptive multitype contrastive views generation for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605513.

[4] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[5] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified *k*-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jul. 2008.

[6] P. O. Gislason, J. A. Benediktsson, and J. R. Sveinsson, "Random forests for land cover classification," *Pattern Recognit. Lett.*, vol. 27, no. 4, pp. 294–300, Mar. 2006.

[7] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[8] C. Shi, P. Chen, L. Fang, M. Zhao, X. Hei, and Q. Miao, "Multiscale spatial–frequency-domain cross-transformer for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–15, 2025.

[9] C. Shi, L. Lu, M. Zhao, X. Hei, C.-M. Pun, and Q. Miao, "Learning hyperspectral noisy label with global and local hypergraph Laplacian energy," *Pattern Recognit.*, vol. 165, Sep. 2025, Art. no. 111606.

[10] C. Shi and C.-M. Pun, "Multiscale superpixel-based hyperspectral image classification using recurrent neural networks with stacked autoencoders," *IEEE Trans. Multimedia*, vol. 22, no. 2, pp. 487–501, Feb. 2020.

[11] R. Wang, Z. Wu, Z. Weng, J. Chen, G.-J. Qi, and Y.-G. Jiang, "Cross-domain contrastive learning for unsupervised domain adaptation," *IEEE Trans. Multimedia*, vol. 25, pp. 1665–1673, 2023.

[12] J. Yuan et al., "Domain-aware graph network for bridging multi-source domain adaptation," *IEEE Trans. Multimedia*, vol. 26, pp. 7210–7224, 2024.

[13] P. He, L. Jiao, F. Liu, X. Liu, R. Shang, and S. Wang, "Cross-domain scene unsupervised learning segmentation with dynamic subdomains," *IEEE Trans. Multimedia*, vol. 26, pp. 6770–6784, 2024.

[14] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2022.

[15] J. Bai et al., "Few-shot hyperspectral image classification based on adaptive subspaces and feature transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523917.

[16] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[17] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[18] Z. Li et al., "Few-shot hyperspectral image classification with self-supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517917.

[19] R. Shang, Y. Xie, W. Zhang, J. Feng, and S. Xu, "Joint adversarial network with semantic and topology fusion for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521116.

[20] Q. Liu, J. Peng, N. Chen, W. Sun, Y. Ning, and Q. Du, "Category-specific prototype self-refinement contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524416.

[21] F. Liu, W. Gao, J. Liu, X. Tang, and L. Xiao, "Adversarial domain alignment with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525720.

[22] Z. Wang, S. Zhao, G. Zhao, and X. Song, "Dual-branch domain adaptation few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506116.

[23] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.

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[24] X. Zhang et al., "Spectral–spatial distribution consistent network based on meta-learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520915.
[25] L. Hu, W. He, L. Zhang, and H. Zhang, "Cross-domain meta-learning under dual-adjustment mode for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526416.
[26] H. Liu, J. He, Y. Li, and Y. Bi, "Multilevel prototype alignment for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4400115.
[27] Z. Ye, J. Wang, T. Sun, J. Zhang, and W. Li, "Cross-domain few-shot learning based on graph convolution contrast for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504614.
[28] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.
[29] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.
[30] W. Guo, H. Ye, and F. Cao, "Feature-grouped network with spectral–spatial connected attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5500413.
[31] P. Duan, P. Ghamisi, X. Kang, B. Rasti, S. Li, and R. Gloaguen, "Fusion of dual spatial information for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7726–7738, Sep. 2021.
[32] H. Chen, F. Miao, Y. Chen, Y. Xiong, and T. Chen, "A hyperspectral image classification method using multifeature vectors and optimized KELM," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2781–2795, 2021.
[33] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[34] J.-Y. Yang, H.-C. Li, W.-S. Hu, L. Pan, and Q. Du, "Adaptive cross-attention-driven spatial–spectral graph convolutional network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[35] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.
[36] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.
[37] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.
[38] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.
[39] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.
[40] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511614.
[41] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
[42] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.
[43] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.
[44] M. Zhang, H. Liu, M. Gong, H. Li, Y. Wu, and X. Jiang, "Cross-domain self-taught network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4501719.
[45] C. Liu et al., "Multi-level relation learning for cross-domain few-shot hyperspectral image classification," *Int. J. Speech Technol.*, vol. 54, no. 5, pp. 4392–4410, Mar. 2024.
[46] J. Huang, D. Guan, A. Xiao, S. Lu, and L. Shao, "Category contrast for unsupervised domain adaptation in visual tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, Jun. 2022, pp. 1193–1204.
[47] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.
[48] M. Ye, X. Zhang, P. C. Yuen, and S.-F. Chang, "Unsupervised embedding learning via invariant and spreading instance feature," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6203–6212.
[49] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.
[50] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
[51] P. Khosla et al., "Supervised contrastive learning," in *Proc. NIPS*, 2020, pp. 18661–18673.
[52] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 30, Jul. 2017, pp. 1–11.
[53] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.
[54] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.
[55] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[56] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[57] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.
[58] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.
[59] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[60] F. Iandola, M. Moskewicz, S. Karayev, R. Girshick, T. Darrell, and K. Keutzer, "DenseNet: Implementing efficient ConvNet descriptor pyramids," 2014, *arXiv:1404.1869*.
[61] Z. Xue, Z. Liu, and M. Zhang, "DSR-GCN: Differentiated-scale restricted graph convolutional network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504918.
[62] Z. Han et al., "Multisource collaborative domain generalization for cross-scene remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535815.



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